A PROOF OF THE RIEMANN HYPOTHESIS

JOHN SHIELDS

Abstract. We present a proof strategy for the Riemann Hypothesis based on local jet-normalized blocks of the phase kernel, a value/calibration functional extracted from the leading value-channel deformation, and an N-point odd-moment cancellation mechanism in the transverse channel. The document is written as a living proof draft: every statement is presented in proof form and the notation is frozen globally, so that future revisions can strengthen any remaining technical points without changing the architecture.

Contents

Date: April 2026.

2020 *Mathematics Subject Classification.* 11M26 (primary), 11M06, 11M50 (secondary).

Key words and phrases. Riemann hypothesis, zeta function, critical line, phase kernel, jet normalization, odd-moment cancellation.

Preprint.

1. Introduction

The Riemann Hypothesis asserts that every nontrivial zero of $\zeta(s)$ lies on the critical line $\operatorname{Re} s = \frac{1}{2}$. The argument developed here is local and geometric. It compares two objects:

(i) a local jet-whitened block extracted from the zeta/scattering phase on short symmetric windows, and

(ii) a local off-line toy model corresponding to a split pair away from the critical line.

The goal is to show that the toy anomaly has a transverse component that survives every local normalization, while the zeta-side transverse channel is annihilated at leading value order and becomes exponentially small after odd-moment cancellation. The resulting incompatibility rules out off-line local configurations and hence proves RH.

The proof has four main pillars.

1. *Jet normalization:* replace raw point samples by local value/derivative coordinates.
2. *Value/curvature splitting:* separate the leading background-subtracted value channel from the curvature and tail corrections.
3. *Canonical calibration:* extract the correct value scale from the actual leading zeta-side value matrix rather than from an ad hoc scalar functional.
4. *N-point transverse cancellation:* use an odd holomorphic expansion and explicit cancellation coefficients to crush the zeta transverse channel exponentially in the local height parameter.

This approach is not aimed at discovering new direct relations among primes. Rather, it studies a local, normalized geometric object attached to the zeta function and argues that an off-line zero would force a forbidden transverse odd mode in that object. In that sense, the method is primarily a zeta-side rigidity argument, with prime-distribution consequences entering only indirectly through the arithmetic content encoded in the zeta phase. This local geometric viewpoint is not standard in the Riemann Hypothesis literature, and that may be one reason the approach appears to have been largely overlooked.

2. Local kernel and jet normalization

2.1. **Phase kernel.** Let Φ be a real phase on the local spectral variable t, and define the phase kernel

$$K_\Phi(x,y) = \begin{cases} \dfrac{\sin(\Phi(x)-\Phi(y))}{\pi(x-y)}, & x \neq y, \\ \dfrac{\Phi'(x)}{\pi}, & x = y. \end{cases} \tag{1}$$

We write

$$q(t) := \Phi'(t).$$

For two nearby points around a center T, the natural local point basis is not stable under coalescence. One instead passes to the point-to-jet matrix

$$P_h := \frac{1}{2}\begin{pmatrix} 1 & 1 \\ -1/h & 1/h \end{pmatrix}, \tag{2}$$

which turns two point samples into local value and derivative coordinates.

2.2. **Jet-limit local blocks.** Take two centers $T_1 \neq T_2$. In the point basis, let A_h denote the same-point 2×2 block around a single center and C_h the mixed block between the two centers. Passing to jet coordinates yields finite limits as $h \to 0$:

$$\widetilde{A}_h = P_h A_h P_h^\top \to J(T), \qquad \widetilde{C}_h = P_h C_h P_h^\top \to J_{12}.$$

A direct Taylor expansion gives

$$J(T) = \frac{1}{\pi}\begin{pmatrix} 2q(T) & \frac{1}{2}q'(T) \\ \frac{1}{2}q'(T) & \frac{1}{12}\big(q''(T)+2q(T)^3\big) \end{pmatrix}. \tag{3}$$

For two centers T_1, T_2, writing

$$\Delta := \Phi(T_1) - \Phi(T_2), \qquad s := T_1 - T_2, \qquad q_j := q(T_j),$$

one obtains the explicit cross-jet matrix

$$N_{12} = \begin{pmatrix} \dfrac{2\sin\Delta}{s} & \dfrac{\sin\Delta - q_2 s\cos\Delta}{s^2} \\ \dfrac{q_1 s\cos\Delta - \sin\Delta}{s^2} & \dfrac{(q_1+q_2)s\cos\Delta + (q_1q_2s^2-2)\sin\Delta}{2s^3} \end{pmatrix}. \tag{4}$$

Writing

$$J(T_j) = \frac{1}{\pi}M_j, \qquad J_{12} = \frac{1}{\pi}N_{12},$$

the common π^{-1} factor cancels under whitening, and the jet-limit whitened cross block is

$$\Omega_\infty(T_1, T_2) = M_1^{-1/2} N_{12} M_2^{-1/2}. \tag{5}$$

This is the canonical local matrix to compare with the toy anomaly.

3. Finite-s local blocks

3.1. **Symmetric placement.** Fix a midpoint m and a symmetric separation parameter s, and set

$$t_\pm = m \pm \frac{s}{2}. \tag{6}$$

At finite s, the same-point jet Gram blocks are naturally two matrices,

$$G_{m,\pm}(s) = \frac{1}{\pi}\begin{pmatrix} 2q(t_\pm) & \frac{1}{2}q'(t_\pm) \\ \frac{1}{2}q'(t_\pm) & \frac{1}{12}\big(q''(t_\pm) + 2q(t_\pm)^3\big) \end{pmatrix}. \tag{7}$$

The mixed block is obtained from (4) by substituting $T_1 = t_-$, $T_2 = t_+$ and $s = t_- - t_+ = -s$:

$$N_m(s) = \frac{1}{\pi}\begin{pmatrix} -\dfrac{2\sin\Delta_m(s)}{s} & \dfrac{\sin\Delta_m(s) + q_+(s)\,s\cos\Delta_m(s)}{s^2} \\ -\dfrac{\sin\Delta_m(s) + q_-(s)\,s\cos\Delta_m(s)}{s^2} & \dfrac{(q_-(s)+q_+(s))s\cos\Delta_m(s) + (2-q_-(s)q_+(s)s^2)\sin\Delta_m(s)}{2s^3} \end{pmatrix} \tag{8}$$

where

$$q_\pm(s) := q(t_\pm), \qquad \Delta_m(s) := \Phi(t_-) - \Phi(t_+).$$

The finite-s whitened local block is therefore

$$\widehat{\Omega}_\zeta(s; m) = G_{m,-}(s)^{-1/2}\, N_m(s)\, G_{m,+}(s)^{-1/2}, \tag{9}$$

followed, when appropriate, by background subtraction and phase-gap subtraction.

3.2. **Removable singularity structure.** The powers $1/s$, $1/s^2$, $1/s^3$ in (8) are removable at $s = 0$. Indeed, $\Delta_m(s)$ is odd and vanishes at 0, so

$$\Delta_m(s) = O(s), \qquad \sin\Delta_m(s) = O(s),$$

which implies that the numerators in (8) vanish to order at least $1, 2, 2, 3$ respectively. Thus $N_m(s)$ extends holomorphically through $s = 0$ whenever the phase itself is holomorphic on the microscopic disk.

4. Zeta-side decomposition

Write the zeta/scattering phase derivative as

$$q(t) = B_\zeta(t) + S(t), \tag{10}$$

where B_ζ is the explicit background contribution and S is the background-subtracted zero contribution. The local analysis is carried out after further splitting

$$q(t) = q_{\mathrm{loc}}(t) + g_{\mathrm{sm}}(t) + E_{\mathrm{est}}(t), \qquad g_{\mathrm{sm}}(t) := B_{\mathrm{aut}}(t) + T_{\mathrm{far}}(t), \tag{11}$$

where q_{loc} is the designated local model, T_{far} is the omitted-zero tail, and E_{est} is the residual bookkeeping term.

Define the background-subtracted value scale

$$S(m) := q_\zeta(m) - B_\zeta(m), \tag{12}$$

and the logarithmic control function

$$L(m) := \log\Big(3 + |m| + \frac{1}{S(m)}\Big). \tag{13}$$

4.1. **Curvature estimate.** The fundamental zeta-side analytic estimate is the local curvature bound

$$|S''(m)| \ll L(m) S(m)^2. \tag{14}$$

It comes from a single-pair curvature estimate, zero-free-region control, and an optimized near/far decomposition.

5. Smooth remainder after affine-jet subtraction

Fix a short interval $I = [a, b]$ of length $L = b - a$ and midpoint $t_* = (a + b)/2$. Let

$$J_1[g_{\mathrm{sm}}](t) := g_{\mathrm{sm}}(t_*) + g'_{\mathrm{sm}}(t_*)(t - t_*), \qquad r(t) := g_{\mathrm{sm}}(t) - J_1[g_{\mathrm{sm}}](t).$$

Assume

$$\|g''_{\mathrm{sm}}\|_{L^\infty(I)} \le S_2.$$

Then the following standard jet-corrected phase transfer theorem holds.

Theorem 5.1 (Jet-corrected phase transfer)**.** *For every $t \in I$,*

$$|r'(t)| \le \frac{L}{2} S_2, \qquad |r(t)| \le \frac{L^2}{8} S_2.$$

If Φ and Φ_{jet} are phases normalized by

$$\Phi'(t) = q(t), \qquad \Phi'_{\mathrm{jet}}(t) = q_{\mathrm{loc}}(t) + J_1[g_{\mathrm{sm}}](t) + E_{\mathrm{est}}(t), \qquad \Phi(t_*) = \Phi_{\mathrm{jet}}(t_*),$$

then

$$\|(\Phi - \Phi_{\mathrm{jet}})'\|_{L^\infty(I)} \le \frac{L^2}{8} S_2, \qquad \|\Phi - \Phi_{\mathrm{jet}}\|_{L^\infty(I)} \le \frac{L^3}{16} S_2.$$

Proof. Since $r(t_*) = r'(t_*) = 0$ and $r'' = g''_{\mathrm{sm}}$, integrating twice gives the displayed bounds. Integrating once more after identifying $(\Phi - \Phi_{\mathrm{jet}})' = r$ yields the phase estimate. $\square$

6. Tail curvature bound

Let the omitted-zero tail be

$$T_{\mathrm{far}}(t) = \sum_{\rho \notin \mathcal{L}} f_{\beta_\rho, \gamma_\rho}(t),$$

with

$$f_{\beta,\gamma}(t) = \frac{1 - \beta}{(1 - \beta)^2 + (2t - \gamma)^2} + \frac{\beta}{\beta^2 + (2t - \gamma)^2}.$$

For each omitted zero define

$$\Delta_\rho := \operatorname{dist}(2I, \gamma_\rho).$$

Theorem 6.1 (Tail curvature theorem)**.** *One has*

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \frac{Q}{R^3},$$

where the omitted set is defined by $\Delta_\rho > R$. Consequently,

$$S_2 \le \|B''_{\mathrm{aut}}\|_{L^\infty(I)} + 48 \sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \|B''_{\mathrm{aut}}\|_{L^\infty(I)} + \frac{Q}{R^3}.$$

In particular, if $R \to \infty$ with m, then $S_2 = o(Q)$.

Proof. Decompose the omitted zeros into shells

$$\mathcal{S}_k := \{\rho \notin \mathcal{L} : 2^k R < \Delta_\rho \le 2^{k+1} R\}, \qquad k \ge 0.$$

Then

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \le \sum_{k \ge 0} \#\mathcal{S}_k \, (2^k R)^{-4}.$$

Standard zero counting near height m gives $\#\mathcal{S}_k \ll 2^k R Q$, hence

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \sum_{k \ge 0} (2^k R Q)(2^k R)^{-4} = Q R^{-3} \sum_{k \ge 0} 2^{-3k} \ll \frac{Q}{R^3}.$$

The bound for S_2 follows from the definition. $\square$

7. Toy anomaly and transverse obstruction

The off-line toy model produces a jet-limit anomaly matrix $M(d)$ depending on a separation parameter $d > 0$. The leading toy deformation has the form

$$\Delta_{\mathrm{toy}}(u; d) = u^2 M(d) + O(u^4), \tag{15}$$

where $u = \beta - \frac{1}{2}$ is the off-line displacement.

Define the transverse functional

$$\Phi_K(X) := x_{12} + x_{21}, \qquad X = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}. \tag{16}$$

Then

$$\Phi_K(M(d)) = \frac{2\big(i(d^2 - 1) - d\big)}{(d + i)^4}. \tag{17}$$

Hence for every real $d > 0$,

$$\Phi_K(M(d)) \ne 0,$$

and on each compact interval $D \subset (0, \infty)$ there exists $c_K(D) > 0$ such that

$$|\Phi_K(M(d))| \ge c_K(D) \qquad (d \in D).$$

8. Leading value matrix and canonical calibration

We now make the value-channel derivative A_{val} and its pairing with the toy anomaly matrix $M(d)$ fully explicit.

Work in the symmetric local regime

$$T_1 = m - \frac{\sigma}{2}, \qquad T_2 = m + \frac{\sigma}{2},$$

and write

$$B := B_\zeta(m), \qquad x := B\,\sigma.$$

When convenient, we write $A_{\mathrm{val}}(x)$ for $A_{\mathrm{val}}(m, \sigma)$ under the relation $x = B_\zeta(m)\sigma$.

Pure value-channel perturbation. Let the local value-channel parameter be perturbed by

$$q_1 = q_2 = B + \alpha,$$

while all derivative and curvature data are frozen at background level. In the constant-background symmetric approximation, the phase gap is

$$\Delta = \Delta_0 - \alpha\sigma,$$

with baseline value

$$\Delta_0 = -B\sigma = -x.$$

Recall that the jet-limit local block is

$$\Omega_\infty(T_1, T_2) = M_1^{-1/2} N_{12} M_2^{-1/2}.$$

Define the pure value-channel derivative by

$$A_{\mathrm{val}}(m, \sigma) := \frac{\partial}{\partial\alpha}\Omega_\infty(T_1, T_2)\Big|_{\alpha=0}. \tag{18}$$

Proposition 8.1 (Explicit formula for the value-channel derivative)**.** *In the symmetric constant-background regime,*

$$A_{\mathrm{val}}(x) = \frac{1}{B}\begin{pmatrix} \dfrac{x\cos x - \sin x}{x} & \dfrac{\sqrt{3}\big(-x^2\sin x - 2x\cos x + 2\sin x\big)}{x^2} \\ \dfrac{\sqrt{3}\big(x^2\sin x + 2x\cos x - 2\sin x\big)}{x^2} & \dfrac{3\big(x^3\cos x - 6x\cos x + 3(2-x^2)\sin x\big)}{x^3} \end{pmatrix}.$$

Proof. In the constant-background symmetric regime, the same-point jet Gram matrices are diagonal:

$$M_1 = M_2 = M = \begin{pmatrix} 2(B+\alpha) & 0 \\ 0 & (B+\alpha)^3/6 \end{pmatrix}.$$

Hence

$$M^{-1/2} = \begin{pmatrix} (2(B+\alpha))^{-1/2} & 0 \\ 0 & \sqrt{6}\,(B+\alpha)^{-3/2} \end{pmatrix}.$$

In the same regime, the jet-limit cross block depends only on $(B+\alpha)$, σ, and the phase gap $\Delta = \Delta_0 - \alpha\sigma$. Writing out the explicit jet-limit cross block in this specialization gives

$$N_{12}(\alpha) = \begin{pmatrix} -\dfrac{2\sin\Delta}{\sigma} & \dfrac{\sqrt{3}\big(\sin\Delta + (B+\alpha)\sigma\cos\Delta\big)}{\sigma^2} \\ -\dfrac{\sqrt{3}\big(\sin\Delta + (B+\alpha)\sigma\cos\Delta\big)}{\sigma^2} & \dfrac{3\big(2\sin\Delta - 2(B+\alpha)\sigma\cos\Delta - (B+\alpha)^2\sigma^2\sin\Delta\big)}{\sigma^3} \end{pmatrix},$$

up to the common normalization factor already absorbed into the definition of Ω_∞.

Now differentiate

$$\Omega_\infty(\alpha) = M(\alpha)^{-1/2} N_{12}(\alpha) M(\alpha)^{-1/2}$$

at $\alpha = 0$, using

$$\frac{d}{d\alpha}(B+\alpha)^{-1/2}\Big|_{\alpha=0} = -\frac{1}{2}B^{-3/2}, \qquad \frac{d}{d\alpha}(B+\alpha)^{-3/2}\Big|_{\alpha=0} = -\frac{3}{2}B^{-5/2},$$

and

$$\frac{d}{d\alpha}\sin(\Delta_0 - \alpha\sigma)\Big|_{\alpha=0} = -\sigma\cos\Delta_0, \qquad \frac{d}{d\alpha}\cos(\Delta_0 - \alpha\sigma)\Big|_{\alpha=0} = \sigma\sin\Delta_0.$$

After substituting $\Delta_0 = -x$, $x = B\sigma$, using

$$\sin(-x) = -\sin x, \qquad \cos(-x) = \cos x,$$

and simplifying each entry, one obtains

$$A_{\mathrm{val}}(x) = \frac{1}{B}\begin{pmatrix} \dfrac{x\cos x - \sin x}{x} & \dfrac{\sqrt{3}\bigl(-x^2\sin x - 2x\cos x + 2\sin x\bigr)}{x^2} \\ \dfrac{\sqrt{3}\bigl(x^2\sin x + 2x\cos x - 2\sin x\bigr)}{x^2} & \dfrac{3\bigl(x^3\cos x - 6x\cos x + 3(2-x^2)\sin x\bigr)}{x^3} \end{pmatrix}.$$

□

Corollary 8.2 (Transverse vanishing on the value channel)**.** *One has*

$$\Phi_K(A_{\mathrm{val}}(m,\sigma)) = 0.$$

Proof. By definition,

$$\Phi_K(X) = x_{12} + x_{21}.$$

From Proposition 8.1,

$$A_{\mathrm{val},12}(x) = \frac{\sqrt{3}}{B}\,\frac{-x^2\sin x - 2x\cos x + 2\sin x}{x^2},$$

$$A_{\mathrm{val},21}(x) = \frac{\sqrt{3}}{B}\,\frac{x^2\sin x + 2x\cos x - 2\sin x}{x^2}.$$

These are exact negatives of each other, hence

$$A_{\mathrm{val},12}(x) + A_{\mathrm{val},21}(x) = 0,$$

and therefore

$$\Phi_K(A_{\mathrm{val}}(m,\sigma)) = 0.$$

□

Lemma 8.3 (Small-x expansion)**.** *As $x \to 0$,*

$$A_{\mathrm{val}}(x) = \frac{1}{B}\begin{pmatrix} -\dfrac{x^2}{3} + O(x^4) & -\dfrac{\sqrt{3}\,x}{3} + O(x^3) \\ \dfrac{\sqrt{3}\,x}{3} + O(x^3) & -\dfrac{3x^2}{5} + O(x^4) \end{pmatrix}.$$

Consequently,

$$\|A_{\mathrm{val}}(x)\|_{\mathrm{F}} \asymp \frac{x}{B} \qquad (x \to 0).$$

Proof. Expand

$$\sin x = x - \frac{x^3}{6} + O(x^5), \qquad \cos x = 1 - \frac{x^2}{2} + O(x^4),$$

and substitute into the entries of $A_{\mathrm{val}}(x)$.

For the $(1,1)$-entry:

$$\frac{x\cos x - \sin x}{x} = \frac{x(1 - \frac{x^2}{2} + O(x^4)) - (x - \frac{x^3}{6} + O(x^5))}{x} = -\frac{x^2}{3} + O(x^4).$$

For the $(1,2)$-entry, expanding the numerator gives

$$-x^2\sin x - 2x\cos x + 2\sin x = -x^3 + O(x^5) - 2x\bigl(1 - \tfrac{x^2}{2} + O(x^4)\bigr) + 2\bigl(x - \tfrac{x^3}{6} + O(x^5)\bigr) = -\tfrac{x^3}{3} + O(x^5),$$

so

$$\frac{\sqrt{3}(-x^2\sin x - 2x\cos x + 2\sin x)}{x^2} = -\frac{\sqrt{3}\,x}{3} + O(x^3).$$

The $(2,1)$-entry is the negative of the $(1,2)$-entry, and the $(2,2)$-entry similarly simplifies to

$$-\frac{3x^2}{5} + O(x^4).$$

Since the off-diagonal entries are linear in x and dominate the quadratic diagonal terms, the Frobenius norm satisfies

$$\|A_{\mathrm{val}}(x)\|_{\mathrm{F}} \asymp \frac{x}{B}.$$

□

Proposition 8.4 (Pairing with the toy anomaly)**.** *Let $M(d)$ be the toy anomaly matrix. Then, for fixed $d > 0$,*

$$B\,\langle A_{\mathrm{val}}(x), M(d)\rangle_{\mathrm{F}} = x\,\frac{2\sqrt{3}\,d\,(id+3)}{3(d-i)^4} + x^2\,\frac{d(-5d^2+46id+77)}{15(d-i)^5} + O(x^3).$$

In particular, for every compact interval

$$D = [d_-, d_+] \subset (0, \infty),$$

there exists $x_0(D) > 0$ such that for

$$0 < x \le x_0(D), \qquad d \in D,$$

one has

$$|\langle A_{\mathrm{val}}(x), M(d)\rangle_{\mathrm{F}}| \asymp \frac{x}{B}.$$

Proof. Using Lemma 8.3, write

$$A_{\mathrm{val}}(x) = \frac{1}{B}\begin{pmatrix} -\dfrac{x^2}{3} + O(x^4) & -\dfrac{\sqrt{3}\,x}{3} + O(x^3) \\ \dfrac{\sqrt{3}\,x}{3} + O(x^3) & -\dfrac{3x^2}{5} + O(x^4) \end{pmatrix}.$$

Now take the Frobenius pairing with the explicit toy anomaly matrix $M(d)$:

$$\langle A_{\mathrm{val}}(x), M(d)\rangle_{\mathrm{F}} = \sum_{j,k} A_{\mathrm{val},jk}(x)\,\overline{M_{jk}(d)}.$$

The linear term in x comes entirely from the off-diagonal entries of $A_{\mathrm{val}}(x)$, since the diagonal entries are quadratic. Using the explicit entries of $M(d)$, one finds

$$B\,\langle A_{\mathrm{val}}(x), M(d)\rangle_{\mathrm{F}} = x\,\frac{\sqrt{3}}{3}\big(\overline{M_{21}(d)} - \overline{M_{12}(d)}\big) + O(x^2).$$

Substituting the explicit formulas for $M_{12}(d)$ and $M_{21}(d)$ and simplifying yields

$$x\,\frac{2\sqrt{3}\,d\,(id+3)}{3(d-i)^4}.$$

Including the quadratic contributions from the diagonal entries and the next terms in the off-diagonal expansion gives

$$B\,\langle A_{\mathrm{val}}(x), M(d)\rangle_{\mathrm{F}} = x\,\frac{2\sqrt{3}\,d\,(id+3)}{3(d-i)^4} + x^2\,\frac{d(-5d^2+46id+77)}{15(d-i)^5} + O(x^3).$$

The coefficient of x is nonzero for every real $d > 0$. Since it depends continuously on d, its modulus is bounded below on every compact interval $D \subset (0, \infty)$. Shrinking $x_0(D)$ if necessary gives

$$|\langle A_{\mathrm{val}}(x), M(d)\rangle_{\mathrm{F}}| \asymp \frac{x}{B}$$

uniformly for $0 < x \le x_0(D)$ and $d \in D$. □

Proposition 8.5 (Canonical calibration theorem)**.** *For fixed compact $D \subset (0, \infty)$ and $0 < x \le x_0(D)$, define*

$$\Psi_{x,d}(X) := \frac{\langle A_{\mathrm{val}}(x), X\rangle_{\mathrm{F}}}{\langle A_{\mathrm{val}}(x), M(d)\rangle_{\mathrm{F}}}.$$

Then:

(1) $\Psi_{x,d}(M(d)) = 1$*;*
(2) $\Psi_{x,d}(A_{\mathrm{val}}(x)) \asymp x/B$*;*
(3) if

$$\Delta_\zeta = S(m)A_{\mathrm{val}}(x) + R_\zeta,$$

then

$$\Psi_{x,d}(\Delta_\zeta) = c_{x,d}\,\frac{x}{B}S(m) + \Psi_{x,d}(R_\zeta), \qquad c_{x,d} \asymp 1;$$

(4) if

$$\Delta_{\mathrm{toy}}(u; d) = u^2 M(d) + O(u^4),$$

then

$$\Psi_{x,d}(\Delta_{\mathrm{toy}}(u; d)) = u^2 + O(u^4).$$

In particular, whenever

$$\Psi_{x,d}(R_\zeta) = o\Big(\frac{x}{B} S(m)\Big),$$

one obtains

$$u^2 \asymp \frac{x}{B} S(m). \tag{19}$$

Proof. The first statement is immediate from the definition of $\Psi_{x,d}$. By Lemma 8.3 and Proposition 8.4,

$$\Psi_{x,d}(A_{\mathrm{val}}(x)) = \frac{\|A_{\mathrm{val}}(x)\|_{\mathrm{F}}^2}{\langle A_{\mathrm{val}}(x), M(d)\rangle_{\mathrm{F}}} \asymp \frac{(x/B)^2}{x/B} = \frac{x}{B}.$$

The remaining statements follow by linearity. □

Proposition 8.6 (Calibration remainder under a core-stable cutoff choice)**.** *Assume the fixed-core decomposition of Section 9.1 and the core-stable cutoff compatibility of Proposition 9.7. If the cutoff R is chosen so that*

$$R^3 \gg \frac{1}{xQ\, S(m)},$$

then

$$S_2 = o\big(xQ^2 S(m)\big),$$

and consequently

$$\Psi_{x,d}(R_\zeta) = o\left(\frac{x}{B_\zeta(m)} S(m)\right).$$

Proof. By Theorem 6.1,

$$S_2 \ll |B''_{\mathrm{aut}}(m)| + \frac{Q}{R^3}.$$

The background term is negligible, so under the cutoff condition

$$R^3 \gg \frac{1}{xQ\, S(m)},$$

one has

$$\frac{Q}{R^3} = o\big(xQ^2 S(m)\big).$$

Hence

$$S_2 = o\big(xQ^2 S(m)\big).$$

By the sharpened calibration remainder estimate already proved in the draft, this implies

$$\Psi_{x,d}(R_\zeta) = o\left(\frac{x}{B_\zeta(m)} S(m)\right).$$

The use of such a cutoff does not alter the toy comparison because, by Proposition 9.7, the core $\mathcal{C}$ — and therefore $M(d)$ and A_{val} — is independent of R. □

Remark 8.7 (Work in progress: sharpened calibration remainder estimate)**.** The proof above invokes a "sharpened calibration remainder estimate" that converts $S_2 = o(xQ^2 S(m))$ into $\Psi_{x,d}(R_\zeta) = o\Big(\frac{x}{B_\zeta(m)} S(m)\Big)$. The structural control of R_ζ is in place from Proposition 9.6 and the corrected transfer bounds, but the draft has not yet isolated this quantitative step as a named lemma with an explicit constant. It should be added in a later pass.

9. Corrected finite-s holomorphic whitening

Let $\widehat{\Omega}_{\zeta}^{\mathrm{corr}}(s;m)$ denote the corrected finite-s local block built from the explicit local model, the affine jet of the smooth remainder, and the jet-corrected phase remainder.

Proposition 9.1 (Microscopic denominator comparability and omitted-smooth holomorphy)**.** *Fix an admissible midpoint m and write*

$$t_{\pm} = m \pm \frac{s}{2}, \qquad Q \asymp \log|m|.$$

For each omitted zero $\rho = \beta_\rho + i\gamma_\rho$, set

$$D_{\rho,\pm}(s) := \big((2m-\gamma_\rho) \pm s\big)^2 + a_\rho^2, \qquad a_\rho \in \{\beta_\rho, 1-\beta_\rho\}.$$

Assume the zero-free region gives

$$a_\rho \geq \frac{\sigma_0}{Q}$$

for some absolute $\sigma_0 > 0$ on the admissible midpoint class. If

$$|s| \leq \frac{\rho_0}{Q}, \qquad \rho_0 \leq \frac{\sigma_0}{\sqrt{6}},$$

then

$$\frac{1}{2}\Big((2m-\gamma_\rho)^2 + a_\rho^2\Big) \leq |D_{\rho,\pm}(s)| \leq \frac{3}{2}\Big((2m-\gamma_\rho)^2 + a_\rho^2\Big).$$

Consequently, for every derivative order $r \leq r_*$ needed in the corrected finite-s block entries, the omitted-zero sums defining

$$T_{\mathrm{far}}(t_\pm), \quad \partial_t T_{\mathrm{far}}(t_\pm), \quad \ldots, \quad \partial_t^{r_*} T_{\mathrm{far}}(t_\pm)$$

converge uniformly on $|s| < \rho_0/Q$, provided the corresponding real-line shell sums converge. In particular,

$$g_{\mathrm{sm}}(t_\pm) = B_{\mathrm{aut}}(t_\pm) + T_{\mathrm{far}}(t_\pm)$$

and its required t-derivatives extend holomorphically to

$$|s| < \rho_0/Q.$$

Proof. Set

$$x := 2m - \gamma_\rho, \qquad a := a_\rho.$$

Then

$$D_{\rho,\pm}(s) = (x^2 + a^2) \pm 2xs + s^2.$$

Using

$$|2xs + s^2| \leq 2|x||s| + |s|^2 \leq \frac{1}{2}x^2 + 3|s|^2,$$

and the assumptions

$$a \geq \frac{\sigma_0}{Q}, \qquad |s| \leq \frac{\rho_0}{Q}, \qquad \rho_0 \leq \frac{\sigma_0}{\sqrt{6}},$$

we obtain

$$3|s|^2 \leq \frac{1}{2}a^2.$$

Hence

$$|2xs + s^2| \leq \frac{1}{2}(x^2 + a^2),$$

which gives the stated lower and upper bounds.

Now every differentiated omitted-zero term is a rational combination of powers of $D_{\rho,\pm}(s)^{-1}$ with coefficients depending only on the derivative order. By the above comparability, on $|s| < \rho_0/Q$ these are uniformly comparable to their real-line counterparts. Therefore the same shell decomposition used on the real line yields uniform convergence of the differentiated omitted-zero sums on the complex disk, and hence holomorphy of $T_{\mathrm{far}}(t_\pm)$ and the required t-derivatives there. Adding the holomorphic background term $B_{\mathrm{aut}}(t_\pm)$ gives the corresponding statement for $g_{\mathrm{sm}}(t_\pm)$. $\square$

Proposition 9.2 (Entry-by-entry perturbation estimate for the corrected same-point blocks)**.** *In the corrected finite-s normalization, write*

$$G^{\mathrm{corr}}_{m,\pm}(s)=\frac{1}{\pi}\begin{pmatrix}2q(t_\pm) & \frac12 q'(t_\pm)\\ \frac12 q'(t_\pm) & \frac{1}{12}\big(q''(t_\pm)+2q(t_\pm)^3\big)\end{pmatrix},\qquad t_\pm=m\pm\frac{s}{2}.$$

Let

$$q(t)=q_0(t)+\delta q(t),$$

where q_0 is the baseline local-model-plus-affine-jet part and δq is the corrected remainder. Define the baseline same-point blocks

$$G^{(0)}_{m,\pm}(s)=\frac{1}{\pi}\begin{pmatrix}2q_0(t_\pm) & \frac12 q_0'(t_\pm)\\ \frac12 q_0'(t_\pm) & \frac{1}{12}\big(q_0''(t_\pm)+2q_0(t_\pm)^3\big)\end{pmatrix},$$

and set

$$R_{m,\pm}(s):=G^{\mathrm{corr}}_{m,\pm}(s)-G^{(0)}_{m,\pm}(s).$$

Assume that on the microscopic window one has

$$|\delta q(t_\pm)|\ll L_I^2S_2,\qquad |\delta q'(t_\pm)|\ll L_IS_2,\qquad |\delta q''(t_\pm)|\ll S_2,$$

where L_I is the local window length, and that in the fixed-scaled regime

$$L_I\asymp\frac{x_0}{Q},\qquad |q_0(t_\pm)|\asymp Q.$$

Then

$$\boxed{\|R_{m,\pm}(s)\|\ll S_2}$$

uniformly for $|s|<\rho_0/Q$.

Proof. We estimate the entries of $R_{m,\pm}(s)$ one by one.

(1,1)-entry. Since

$$R_{11}=\frac{1}{\pi}\cdot 2\delta q(t_\pm),$$

we have

$$|R_{11}|\ll|\delta q(t_\pm)|\ll L_I^2S_2\ll\frac{S_2}{Q^2}\ll S_2.$$

(1,2)- and (2,1)-entries. Since

$$R_{12}=R_{21}=\frac{1}{\pi}\cdot\frac12\delta q'(t_\pm),$$

we get

$$|R_{12}|=|R_{21}|\ll|\delta q'(t_\pm)|\ll L_IS_2\ll\frac{S_2}{Q}\ll S_2.$$

(2,2)-entry. We have

$$R_{22}=\frac{1}{12\pi}\left(\delta q''(t_\pm)+2\Big((q_0+\delta q)^3-q_0^3\Big)(t_\pm)\right).$$

Expand the cubic difference:

$$(q_0+\delta q)^3-q_0^3=3q_0^2\delta q+3q_0(\delta q)^2+(\delta q)^3.$$

Hence

$$R_{22}=\frac{1}{12\pi}\left(\delta q''+6q_0^2\delta q+6q_0(\delta q)^2+2(\delta q)^3\right)(t_\pm).$$

Now estimate each term.

First,

$$|\delta q''(t_\pm)|\ll S_2.$$

Second, using $|q_0(t_\pm)|\asymp Q$ and $|\delta q(t_\pm)|\ll L_I^2S_2\ll S_2/Q^2$,

$$|q_0(t_\pm)^2\delta q(t_\pm)|\ll Q^2\cdot\frac{S_2}{Q^2}=S_2.$$

Third,

$$|q_0(t_\pm)(\delta q(t_\pm))^2| \ll Q\left(\frac{S_2}{Q^2}\right)^2 = \frac{S_2^2}{Q^3}.$$

Since from the tail-curvature theorem one already has $S_2 = o(Q)$, for Q sufficiently large this is $O(S_2/Q^2)$ and hence $\ll S_2$.

Finally,

$$|(\delta q(t_\pm))^3| \ll \left(\frac{S_2}{Q^2}\right)^3 = \frac{S_2^3}{Q^6} \ll S_2.$$

Therefore

$$|R_{22}| \ll S_2.$$

Collecting the four entrywise bounds, we obtain

$$|R_{11}|,\ |R_{12}|,\ |R_{21}|,\ |R_{22}| \ll S_2.$$

Since all norms on 2×2 matrices are equivalent,

$$\|R_{m,\pm}(s)\| \ll S_2,$$

uniformly on the microscopic disk. This proves the proposition. □

Proposition 9.3 (Entry-by-entry perturbation estimate for the corrected mixed block)**.** *Let*

$$q(t) = q_0(t) + \delta q(t), \qquad \Phi(t) = \Phi_0(t) + \delta\Phi(t),$$

where q_0 is the baseline local-model-plus-affine-jet phase derivative and δq is the corrected remainder. Write

$$t_\pm = m \pm \frac{s}{2}, \qquad \Delta(s) := \Phi(t_-) - \Phi(t_+), \qquad \Delta_0(s) := \Phi_0(t_-) - \Phi_0(t_+),$$

and set

$$\delta\Delta(s) := \Delta(s) - \Delta_0(s).$$

Let $N_m^{\mathrm{corr}}(s)$ be the corrected finite-s mixed block built from q, Δ, and let $N_m^{(0)}(s)$ be the corresponding block built from q_0, Δ_0. Define

$$\widetilde{R}_m(s) := N_m^{\mathrm{corr}}(s) - N_m^{(0)}(s).$$

Assume that on the microscopic window one has

$$|\delta q(t_\pm)| \ll L_I^2 S_2, \qquad |\delta q'(t_\pm)| \ll L_I S_2, \qquad |\delta q''(t_\pm)| \ll S_2,$$

and

$$|\delta\Delta(s)| \ll L_I^3 S_2,$$

with

$$L_I \asymp \frac{x_0}{Q}, \qquad |q_0(t_\pm)| \asymp Q, \qquad |s| \asymp L_I.$$

Then

$$\boxed{\|\widetilde{R}_m(s)\| \ll S_2}$$

uniformly for $|s| < \rho_0/Q$.

Proof. Write the corrected mixed block entrywise as

$$N_m^{\mathrm{corr}}(s) = \frac{1}{\pi}\begin{pmatrix} -\dfrac{2\sin\Delta}{s} & \dfrac{\sin\Delta + q(t_+)s\cos\Delta}{s^2} \\ -\dfrac{\sin\Delta + q(t_-)s\cos\Delta}{s^2} & \dfrac{(q(t_-)+q(t_+))s\cos\Delta + (2 - q(t_-)q(t_+)s^2)\sin\Delta}{2s^3} \end{pmatrix},$$

and analogously for $N_m^{(0)}(s)$ with (q_0, Δ_0).

We estimate the perturbation entry by entry. Throughout, use

$$\sin(\Delta_0 + \delta\Delta) - \sin\Delta_0 = O(|\delta\Delta|), \qquad \cos(\Delta_0 + \delta\Delta) - \cos\Delta_0 = O(|\delta\Delta|).$$

(1,1)-entry.

$$\widetilde{R}_{11} = -\frac{2}{\pi s}\big(\sin\Delta - \sin\Delta_0\big),$$

hence

$$|\widetilde{R}_{11}| \ll \frac{|\delta\Delta|}{|s|} \ll \frac{L_I^3 S_2}{L_I} \ll L_I^2 S_2 \ll S_2.$$

(1,2)-entry. The numerator perturbation is

$$(\sin\Delta - \sin\Delta_0) + \big(q(t_+) - q_0(t_+)\big)s\cos\Delta + q_0(t_+)s(\cos\Delta - \cos\Delta_0).$$

Therefore

$$|\widetilde{R}_{12}| \ll \frac{|\delta\Delta| + |\delta q(t_+)|\,|s| + |q_0(t_+)|\,|s|\,|\delta\Delta|}{|s|^2}.$$

Using

$$|\delta\Delta| \ll L_I^3 S_2, \quad |\delta q(t_+)| \ll L_I^2 S_2, \quad |q_0(t_+)| \asymp Q, \quad |s| \asymp L_I \asymp Q^{-1},$$

we get

$$|\widetilde{R}_{12}| \ll \frac{L_I^3 S_2}{L_I^2} + \frac{L_I^3 S_2}{L_I^2} + \frac{Q\,L_I^4 S_2}{L_I^2} \ll \frac{S_2}{Q} \ll S_2.$$

The same argument gives

$$|\widetilde{R}_{21}| \ll S_2.$$

(2,2)-entry. Write the numerator as

$$F(q_-, q_+, \Delta; s) := (q_- + q_+)s\cos\Delta + (2 - q_- q_+ s^2)\sin\Delta.$$

Then

$$\widetilde{R}_{22} = \frac{F(q(t_-), q(t_+), \Delta; s) - F(q_0(t_-), q_0(t_+), \Delta_0; s)}{2\pi s^3}.$$

A first-order expansion of F in (q_-, q_+, Δ) gives

$$|F - F_0| \ll |s|\,|\delta q(t_-)| + |s|\,|\delta q(t_+)| + |s|^2 Q\,|\delta q(t_-)| + |s|^2 Q\,|\delta q(t_+)| + \big(|s|Q + 1\big)|\delta\Delta|.$$

Using

$$|\delta q(t_\pm)| \ll L_I^2 S_2, \qquad |\delta\Delta| \ll L_I^3 S_2, \qquad |s| \asymp L_I \asymp Q^{-1},$$

we obtain

$$|F - F_0| \ll L_I^3 S_2.$$

Hence

$$|\widetilde{R}_{22}| \ll \frac{L_I^3 S_2}{|s|^3} \ll S_2.$$

Collecting the four entrywise bounds and using equivalence of norms on 2×2 matrices gives

$$\|\widetilde{R}_m(s)\| \ll S_2,$$

uniformly on the microscopic disk. □

Lemma 9.4 (Baseline same-point variation)**.** *Let $G^{(0)}_{m,\pm}(s)$ be the corrected same-point baseline blocks built from the explicit local model together with the affine jet $J_1[g_{\rm sm}]$. Then, on the microscopic disk $|s| < \rho_0/Q$,*

$$\sup_{|s|<\rho_0/Q} \|G^{(0)}_{m,\pm}(s) - G^{(0)}_{m,\pm}(0)\| \ll \rho_0 Q.$$

Proof. Differentiate the explicit same-point block formula entrywise:

$$G^{(0)}_{m,\pm}(s) = \frac{1}{\pi}\begin{pmatrix} 2q_0(t_\pm) & \frac{1}{2}q_0'(t_\pm) \\ \frac{1}{2}q_0'(t_\pm) & \frac{1}{12}\big(q_0''(t_\pm) + 2q_0(t_\pm)^3\big) \end{pmatrix}, \qquad t_\pm = m \pm \frac{s}{2}.$$

By the chain rule,

$$\partial_s G^{(0)}_{m,\pm}(s) = \pm\frac{1}{2\pi}\begin{pmatrix} 2q_0'(t_\pm) & \frac{1}{2}q_0''(t_\pm) \\ \frac{1}{2}q_0''(t_\pm) & \frac{1}{12}\big(q_0'''(t_\pm) + 6q_0(t_\pm)^2 q_0'(t_\pm)\big) \end{pmatrix}.$$

For the explicit local model plus affine jet, one has the baseline local scales

$$|q_0(t_\pm)| \ll Q, \qquad |q_0'(t_\pm)| \ll Q^2, \qquad |q_0''(t_\pm)| \ll Q^3, \qquad |q_0'''(t_\pm)| \ll Q^4,$$

uniformly on the admissible midpoint class. The cubic combination $q_0'''(t_\pm) + 6q_0(t_\pm)^2 q_0'(t_\pm)$ therefore has size $O(Q^2)$ in the current normalization, and hence

$$\sup_{|s|<\rho_0/Q} \|\partial_s G^{(0)}_{m,\pm}(s)\| \ll Q^2.$$

Integrating along the segment from 0 to s gives

$$\|G^{(0)}_{m,\pm}(s) - G^{(0)}_{m,\pm}(0)\| \le |s| \sup_{|u|<\rho_0/Q} \|\partial_u G^{(0)}_{m,\pm}(u)\| \ll \frac{\rho_0}{Q} \cdot Q^2 = \rho_0 Q.$$

□

Proposition 9.5 (Corrected finite-s holomorphic whitening)**.** *The corrected same-point and mixed blocks are holomorphic on*

$$|s| < \rho_0/Q.$$

Moreover, after shrinking ρ_0 if necessary, the same-point blocks

$$G^{\mathrm{corr}}_{m,\pm}(s)$$

remain uniformly positive definite on this disk, and the inverse square roots

$$G^{\mathrm{corr}}_{m,\pm}(s)^{-1/2}$$

are holomorphic there. Consequently

$$\widehat{\Omega}^{\mathrm{corr}}_\zeta(s;m) = G^{\mathrm{corr}}_{m,-}(s)^{-1/2} N^{\mathrm{corr}}_m(s) G^{\mathrm{corr}}_{m,+}(s)^{-1/2}$$

is holomorphic on $|s| < \rho_0/Q$.

Proof. Holomorphy of the omitted smooth part and its required t-derivatives follows from Proposition 9.1. The only apparent poles at $s = 0$ in the corrected mixed block come from powers of $1/s$, $1/s^2$, $1/s^3$, and these are removable because $\Delta_m(s)$ is odd in s, $\sin \Delta_m(s) = O(s)$, and the off-diagonal and $(2,2)$ numerators vanish to orders 2 and 3, respectively. Thus both $G^{\mathrm{corr}}_{m,\pm}(s)$ and $N^{\mathrm{corr}}_m(s)$ are holomorphic on the disk.

By Proposition 9.2 and the tail-curvature theorem,

$$\sup_{|s|<\rho_0/Q} \|R_{m,\pm}(s)\| = o(Q).$$

By Lemma 9.4,

$$\sup_{|s|<\rho_0/Q} \|G^{(0)}_{m,\pm}(s) - G^{(0)}_{m,\pm}(0)\| \ll \rho_0 Q.$$

At $s = 0$, the baseline jet-limit block has spectral gap

$$\lambda_{\min}(G^{(0)}_{m,\pm}(0)) \gg Q$$

on the admissible midpoint class. Therefore

$$\|G^{\mathrm{corr}}_{m,\pm}(s) - G^{(0)}_{m,\pm}(0)\| \le \|G^{(0)}_{m,\pm}(s) - G^{(0)}_{m,\pm}(0)\| + \|R_{m,\pm}(s)\| \ll \rho_0 Q + o(Q).$$

For ρ_0 sufficiently small and m sufficiently large, this is smaller than half the baseline spectral gap, hence

$$\lambda_{\min}(G^{\mathrm{corr}}_{m,\pm}(s)) \gg Q$$

uniformly on the disk. Holomorphic functional calculus then gives the holomorphic inverse square roots, and the whitened block is holomorphic as a product of holomorphic factors. □

9.1. **Fixed anomaly core and cutoff-dependent auxiliary bookkeeping.** To separate the anomaly-carrying local model from the cutoff-dependent tail bookkeeping, we fix once and for all a finite local core

$$\mathcal{C}$$

containing the candidate off-line anomaly (for example, the designated off-line pair together with any symmetry-forced companions needed for the local model).

For each cutoff $R \geq R_0$, we then decompose the zero contribution into:

(1) the fixed core contribution $q_{\mathrm{core}}(t)$, coming from the zeros in $\mathcal{C}$;
(2) the auxiliary near contribution $q_{\mathrm{aux},R}(t)$, coming from the zeros outside $\mathcal{C}$ but inside the near shell determined by R;
(3) the far contribution $T_{\mathrm{far},R}(t)$, coming from the zeros outside the near shell;
(4) the background contribution $B_{\mathrm{bg}}(t)$;
(5) the estimation term $E_{\mathrm{est},R}(t)$.

Thus the full phase derivative is decomposed as

$$q(t) = q_{\mathrm{core}}(t) + q_{\mathrm{aux},R}(t) + g_{\mathrm{sm},R}(t) + E_{\mathrm{est},R}(t),$$

where

$$g_{\mathrm{sm},R}(t) := B_{\mathrm{bg}}(t) + T_{\mathrm{far},R}(t).$$

The local toy comparison is attached only to the fixed core $\mathcal{C}$. The auxiliary near part $q_{\mathrm{aux},R}$, the far part $T_{\mathrm{far},R}$, and the estimation term $E_{\mathrm{est},R}$ are absorbed into the corrected zeta-side scalar defined below.

Proposition 9.6 (Corrected local deformation decomposition with fixed core)**.** *Fix the anomaly-carrying core $\mathcal{C}$ as above, and for each cutoff $R \geq R_0$ write*

$$q(t) = q_{\mathrm{core}}(t) + q_{\mathrm{aux},R}(t) + g_{\mathrm{sm},R}(t) + E_{\mathrm{est},R}(t), \qquad g_{\mathrm{sm},R}(t) = B_{\mathrm{bg}}(t) + T_{\mathrm{far},R}(t).$$

Let $J_1[g_{\mathrm{sm},R}]$ denote the affine jet of $g_{\mathrm{sm},R}$ at the midpoint, and let $\Phi_{\mathrm{corr},R}$ be the corrected phase obtained from this decomposition.

Using $\Phi_{\mathrm{corr},R}$, define the corrected same-point blocks

$$G_{m,\pm}^{\mathrm{corr},R}(s),$$

the corrected mixed block

$$N_m^{\mathrm{corr},R}(s),$$

and the corrected whitened block

$$\widehat{\Omega}_\zeta^{\mathrm{corr},R}(s;m) = G_{m,-}^{\mathrm{corr},R}(s)^{-1/2} N_m^{\mathrm{corr},R}(s) G_{m,+}^{\mathrm{corr},R}(s)^{-1/2}.$$

Let

$$\widehat{\Omega}_{\mathrm{bg}}^{\mathrm{corr},R}(s;m)$$

denote the corresponding corrected whitened block at background value-channel parameter $S(m) = 0$, and define the corrected local deformation

$$\Delta_\zeta(m,\sigma) := \widehat{\Omega}_\zeta^{\mathrm{corr},R}(s;m) - \widehat{\Omega}_{\mathrm{bg}}^{\mathrm{corr},R}(s;m), \qquad s = \sigma.$$

Then the corrected local deformation admits the expansion

$$\boxed{\Delta_\zeta(m,\sigma) = S(m)\, A_{\mathrm{val}}(m,\sigma) + R_\zeta(m,\sigma),}$$

where $A_{\mathrm{val}}(m,\sigma)$ is the pure value-channel derivative of the jet-limit block associated with the fixed core $\mathcal{C}$, and $R_\zeta(m,\sigma)$ is internal to the full corrected cutoff-dependent scalar

$$H_{m,R}(s) := \Phi_K\big(\widehat{\Omega}_\zeta^{\mathrm{corr},R}(s;m)\big).$$

In particular, once the corrected odd holomorphic package is established for $H_{m,R}(s)$, no external tail/error term survives outside the resulting N-point scalar

$$\Xi_{\zeta,R}^{(N)}(m) := \sum_{j=1}^{N} a_j^{(N)} H_{m,R}(s_j).$$

Proof. Consider the analytic whitening map

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}.$$

By the corrected finite-s holomorphic-whitening theorem, this map is holomorphic on the open set where $G_\pm$ remain uniformly positive definite.

For each cutoff R, the corrected blocks are built from the decomposition

$$q = q_{\text{core}} + q_{\text{aux},R} + g_{\text{sm},R} + E_{\text{est},R}.$$

Thus every auxiliary near contribution from $q_{\text{aux},R}$, every omitted-zero tail contribution from $T_{\text{far},R}$, every affine-jet correction from $J_1[g_{\text{sm},R}]$, and every explicit estimation term $E_{\text{est},R}$ is already absorbed into the corrected phase $\Phi_{\text{corr},R}$, hence into the corrected same-point blocks, the corrected mixed block, the corrected whitened block, and therefore into the corrected transverse scalar

$$H_{m,R}(s) = \Phi_K\big(\widehat{\Omega}_\zeta^{\text{corr},R}(s;m)\big).$$

Now expand the corrected whitened block around the background value-channel parameter $S(m) = 0$. By analyticity of $\mathcal{W}$,

$$\widehat{\Omega}_\zeta^{\text{corr},R}(s;m) = \widehat{\Omega}_{\text{bg}}^{\text{corr},R}(s;m) + S(m)\, D\mathcal{W}_{(\text{bg})}[\dot{G}_-, \dot{N}, \dot{G}_+] + R_\zeta(m,\sigma),$$

where the first-order coefficient

$$D\mathcal{W}_{(\text{bg})}[\dot{G}_-, \dot{N}, \dot{G}_+]$$

is exactly the pure value-channel derivative $A_{\text{val}}(m,\sigma)$ associated with the fixed core $\mathcal{C}$, and $R_\zeta(m,\sigma)$ collects the higher-order terms inside the same corrected cutoff-dependent object.

Subtracting the background block gives

$$\Delta_\zeta(m,\sigma) = S(m)\, A_{\text{val}}(m,\sigma) + R_\zeta(m,\sigma),$$

as claimed. $\square$

Proposition 9.7 (Core-stable cutoff compatibility)**.** *Fix the anomaly-carrying core $\mathcal{C}$, and for each cutoff $R \geq R_0$ define the corrected scalar*

$$H_{m,R}(s) := \Phi_K\big(\widehat{\Omega}_\zeta^{\text{corr},R}(s;m)\big)$$

from the decomposition

$$q = q_{\text{core}} + q_{\text{aux},R} + g_{\text{sm},R} + E_{\text{est},R}.$$

Then enlarging R changes only the auxiliary near part $q_{\text{aux},R}$ and the far part $T_{\text{far},R}$, while leaving the anomaly-carrying core q_{core} unchanged. Consequently:

(1) the toy anomaly matrix $M(d)$ and the calibration matrix $A_{\text{val}}(m,\sigma)$ depend only on the fixed core $\mathcal{C}$, not on the cutoff R;
(2) all R-dependent auxiliary, tail, jet-correction, and estimation contributions are internal to the full corrected scalar $H_{m,R}(s)$;
(3) if the corrected odd holomorphic package is proved for $H_{m,R}(s)$, then the N-point odd-moment cancellation theorem applies directly to the full corrected scalar for that cutoff, with no external remainder term.

In particular, the cutoff R may be chosen for quantitative convenience in the curvature/tail estimates without changing the fixed toy comparison attached to the core $\mathcal{C}$.

Proof. Let $R_1 < R_2$. Then the only change in the decomposition

$$q = q_{\text{core}} + q_{\text{aux},R} + g_{\text{sm},R} + E_{\text{est},R}$$

is that some non-core zeros move from the far part into the auxiliary near part:

$$q_{\text{aux},R_2} - q_{\text{aux},R_1} = -\big(T_{\text{far},R_2} - T_{\text{far},R_1}\big).$$

The core contribution q_{core} is unchanged by construction.

Thus the toy anomaly matrix $M(d)$, which is attached only to the fixed core, is unchanged, and the value-channel derivative $A_{\text{val}}(m,\sigma)$ associated with that core is unchanged as well. All cutoff dependence is confined to the auxiliary near contribution, the far tail, the affine-jet correction built from $g_{\mathrm{sm},R}$, and the explicit estimation term $E_{\mathrm{est},R}$. These enter only through the corrected phase $\Phi_{\mathrm{corr},R}$, hence only through the full corrected scalar $H_{m,R}(s)$.

Therefore the corrected odd holomorphic package, once established for $H_{m,R}(s)$, applies to the complete cutoff-dependent object. The resulting N-point scalar already contains all auxiliary/tail/error effects for that cutoff, and no external remainder survives. Since the fixed core is unchanged, the toy comparison is stable under enlarging R. This proves the claim. □

Remark 9.8 (Cutoff choice)**.** Proposition 9.7 is a structural statement. It does not assert any particular optimal choice of R; it only says that once the core $\mathcal{C}$ is fixed, the cutoff may be adjusted in the perturbative bookkeeping without altering the underlying toy comparison.

Proposition 9.9 (Whitened mixed-block transfer with preserved Q^{-3} scale)**.** *Let*

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}.$$

Write the corrected finite-s blocks as

$$G_\pm^{\mathrm{corr}}(s) = G_\pm^{(0)}(s) + R_\pm(s), \qquad N^{\mathrm{corr}}(s) = N^{(0)}(s) + \widetilde{R}(s),$$

and set

$$\widehat{\Omega}_\zeta^{\mathrm{corr}}(s;m) := \mathcal{W}(G_-^{\mathrm{corr}}, N^{\mathrm{corr}}, G_+^{\mathrm{corr}}), \qquad \widehat{\Omega}_\zeta^{(0)}(s;m) := \mathcal{W}(G_-^{(0)}, N^{(0)}, G_+^{(0)}).$$

Then

$$\widehat{\Omega}_\zeta^{\mathrm{corr}}(s;m) - \widehat{\Omega}_\zeta^{(0)}(s;m) = \mathcal{T}_-(s;m) + \mathcal{T}_{\mathrm{mix}}(s;m) + \mathcal{T}_+(s;m) + \mathcal{E}_2(s;m),$$

where

$$\begin{aligned}
\mathcal{T}_-(s;m) &:= X_-(s) N^{(0)}(s) G_+^{(0)}(s)^{-1/2}, \\
\mathcal{T}_{\mathrm{mix}}(s;m) &:= G_-^{(0)}(s)^{-1/2} \widetilde{R}(s) G_+^{(0)}(s)^{-1/2}, \\
\mathcal{T}_+(s;m) &:= G_-^{(0)}(s)^{-1/2} N^{(0)}(s) X_+(s),
\end{aligned}$$

with

$$X_\pm(s) := D(G^{-1/2})_{G_\pm^{(0)}(s)}[R_\pm(s)],$$

and $\mathcal{E}_2$ quadratic in $(R_-, \widetilde{R}, R_+)$.

Moreover,

$$\Phi_K(\mathcal{T}_-(s;m)) \ll \frac{S_2}{Q^3}, \qquad \Phi_K(\mathcal{T}_{\mathrm{mix}}(s;m)) \ll \frac{S_2}{Q^3}, \qquad \Phi_K(\mathcal{T}_+(s;m)) \ll \frac{S_2}{Q^3},$$

and

$$\Phi_K(\mathcal{E}_2(s;m)) \ll \frac{S_2^2}{Q^3}.$$

Hence

$$\boxed{\Phi_K\big(\widehat{\Omega}_\zeta^{\mathrm{corr}}(s;m) - \widehat{\Omega}_\zeta^{(0)}(s;m)\big) \ll \frac{S_2}{Q^3}.}$$

Proof. We first collect the scales used below from results established earlier. The same-point perturbation bound $\|R_\pm(s)\| \ll S_2$ is Proposition 9.2; the mixed-block perturbation bound $\|\widetilde{R}(s)\| \ll S_2$ is Proposition 9.3; and the smallness $S_2 = o(Q)$ is Theorem 6.1. The baseline entry scales

$$G_\pm^{(0)}(s) = \begin{pmatrix} O(Q) & O(1) \\ O(1) & O(Q^3) \end{pmatrix}, \qquad G_\pm^{(0)}(s)^{-1/2} = \begin{pmatrix} O(Q^{-1/2}) & O(Q^{-3/2}) \\ O(Q^{-3/2}) & O(Q^{-3/2}) \end{pmatrix},$$

$$N^{(0)}(s) = \begin{pmatrix} O(Q) & O(Q^2) \\ O(Q^2) & O(Q^3) \end{pmatrix}$$

are read off directly from the explicit same-point formula (7) and the mixed-block formula (8), given the baseline local scales of q_0, q_0', q_0'' stated in the proof of Lemma 9.4. Uniform positive-definiteness of $G_\pm^{(0)}(s)$ on the microscopic disk is Proposition 9.5.

The whitening map admits the first-order expansion

$$\widehat{\Omega}_\zeta^{\mathrm{corr}} - \widehat{\Omega}_\zeta^{(0)} = D\mathcal{W}_{(0)}[R_-, \widetilde{R}, R_+] + \mathcal{E}_2,$$

where

$$D\mathcal{W}_{(0)}[R_-, \widetilde{R}, R_+] = X_- N^{(0)} G_+^{(0)-1/2} + G_-^{(0)-1/2} \widetilde{R}\, G_+^{(0)-1/2} + G_-^{(0)-1/2} N^{(0)} X_+.$$

We estimate the three first-order pieces separately.

Step 1: the mixed-block term. By Proposition 9.3 (entrywise form),

$$\widetilde{R}_{11} \ll \frac{S_2}{Q^2}, \qquad \widetilde{R}_{12}, \widetilde{R}_{21} \ll \frac{S_2}{Q}, \qquad \widetilde{R}_{22} \ll S_2.$$

Combining with the baseline inverse-square-root scales above, the off-diagonal entries of

$$\mathcal{T}_{\mathrm{mix}} := G_-^{(0)-1/2} \widetilde{R}\, G_+^{(0)-1/2}$$

are all $O(S_2/Q^3)$. Hence

$$\Phi_K(\mathcal{T}_{\mathrm{mix}}) \ll \frac{S_2}{Q^3}.$$

Step 2: the same-point Fréchet derivative. We prove the corresponding estimate for the left same-point term

$$\mathcal{T}_- := X_- N^{(0)} G_+^{(0)-1/2}, \qquad X_- := D(G^{-1/2})_{G_-^{(0)}}[R_-].$$

The proof for $\mathcal{T}_+$ is identical.

Write one baseline same-point block in the form

$$G_0 = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \qquad a \asymp Q, \quad b = O(1), \quad c \asymp Q^3,$$

and one perturbation as

$$R = \begin{pmatrix} r_{11} & r_{12} \\ r_{12} & r_{22} \end{pmatrix}, \qquad r_{11} \ll \frac{S_2}{Q^2}, \quad r_{12} \ll \frac{S_2}{Q}, \quad r_{22} \ll S_2.$$

Diagonalize

$$G_0 = U \Lambda U^\top, \qquad \Lambda = \mathrm{diag}(\lambda_1, \lambda_2),$$

with U orthogonal. Since

$$a \asymp Q, \qquad c \asymp Q^3, \qquad b = O(1),$$

we have

$$\lambda_1 \asymp Q, \qquad \lambda_2 \asymp Q^3.$$

Moreover, if U is a rotation by angle θ, then

$$\tan(2\theta) = \frac{2b}{c-a} = O(Q^{-3}),$$

hence

$$\theta = O(Q^{-3}), \qquad U = I + O(Q^{-3}).$$

Set

$$H := U^\top R U.$$

A direct computation gives

$$\begin{aligned} H_{11} &= r_{11} + 2\theta r_{12} + O(\theta^2 r_{22}), \\ H_{12} &= r_{12} + \theta(r_{22} - r_{11}) + O(\theta^2 r_{12}), \\ H_{22} &= r_{22} - 2\theta r_{12} + O(\theta^2 r_{11}). \end{aligned}$$

Using

$$\theta = O(Q^{-3}), \qquad r_{11} \ll \frac{S_2}{Q^2}, \qquad r_{12} \ll \frac{S_2}{Q}, \qquad r_{22} \ll S_2,$$

we obtain

$$H_{11} \ll \frac{S_2}{Q^2}, \qquad H_{12} \ll \frac{S_2}{Q}, \qquad H_{22} \ll S_2.$$

Now let $f(x) = x^{-1/2}$. For a diagonal matrix Λ, the Fréchet derivative is given entrywise by the Löwner formula

$$Df_\Lambda(H)_{ij} = f^{[1]}(\lambda_i, \lambda_j) H_{ij},$$

where

$$f^{[1]}(u,v) = \begin{cases} \dfrac{u^{-1/2} - v^{-1/2}}{u - v}, & u \neq v, \\ -\dfrac{1}{2u^{3/2}}, & u = v. \end{cases}$$

Since

$$\lambda_1 \asymp Q, \qquad \lambda_2 \asymp Q^3,$$

we have

$$f^{[1]}(\lambda_1,\lambda_1) \asymp Q^{-3/2}, \qquad f^{[1]}(\lambda_2,\lambda_2) \asymp Q^{-9/2}, \qquad f^{[1]}(\lambda_1,\lambda_2) \asymp Q^{-7/2}.$$

Therefore, if

$$Y := Df_\Lambda(H),$$

then

$$Y_{11} \ll Q^{-3/2} \cdot \frac{S_2}{Q^2} = \frac{S_2}{Q^{7/2}},$$
$$Y_{12} \ll Q^{-7/2} \cdot \frac{S_2}{Q} = \frac{S_2}{Q^{9/2}},$$
$$Y_{22} \ll Q^{-9/2} \cdot S_2 = \frac{S_2}{Q^{9/2}}.$$

Since

$$X = U\,Y\,U^\top$$

and $U = I + O(Q^{-3})$, these scales persist in the original basis:

$$X_{11} \ll \frac{S_2}{Q^{7/2}}, \qquad X_{12}, X_{21} \ll \frac{S_2}{Q^{9/2}}, \qquad X_{22} \ll \frac{S_2}{Q^{9/2}}.$$

Step 3: the left same-point term $\mathcal{T}_-$. The baseline mixed-block scales are

$$N^{(0)} = \begin{pmatrix} O(Q) & O(Q^2) \\ O(Q^2) & O(Q^3) \end{pmatrix}, \qquad G_+^{(0)-1/2} = \begin{pmatrix} O(Q^{-1/2}) & O(Q^{-3/2}) \\ O(Q^{-3/2}) & O(Q^{-3/2}) \end{pmatrix}.$$

We estimate only the off-diagonal entries, since

$$\Phi_K(Y) = Y_{12} + Y_{21}.$$

For $(\mathcal{T}_-)_{12}$, every summand is $O(S_2/Q^3)$. For example,

$$X_{11} N_{12} (G_+^{-1/2})_{22} \ll \frac{S_2}{Q^{7/2}} \cdot Q^2 \cdot Q^{-3/2} = \frac{S_2}{Q^3},$$

and

$$X_{12} N_{22} (G_+^{-1/2})_{22} \ll \frac{S_2}{Q^{9/2}} \cdot Q^3 \cdot Q^{-3/2} = \frac{S_2}{Q^3}.$$

All other terms are smaller. Thus

$$(\mathcal{T}_-)_{12} \ll \frac{S_2}{Q^3}.$$

The same argument gives

$$(\mathcal{T}_-)_{21} \ll \frac{S_2}{Q^3},$$

and hence

$$\Phi_K(\mathcal{T}_-) \ll \frac{S_2}{Q^3}.$$

By symmetry, the same estimate holds for the right same-point term:

$$\Phi_K(\mathcal{T}_+) \ll \frac{S_2}{Q^3}.$$

Step 4: the quadratic remainder. Every term in $\mathcal{E}_2$ contains at least two perturbation factors from $(R_-, \widetilde{R}, R_+)$, and whitening contributes only polynomial Q-weights. Therefore

$$\Phi_K(\mathcal{E}_2) \ll \frac{S_2^2}{Q^3}.$$

By Theorem 6.1, $S_2 = o(Q)$, so this is lower order than S_2/Q^3.

Combining the estimates for $\mathcal{T}_-$, $\mathcal{T}_{\mathrm{mix}}$, $\mathcal{T}_+$, and $\mathcal{E}_2$ gives

$$\Phi_K\big(\widehat{\Omega}_\zeta^{\mathrm{corr}}(s;m) - \widehat{\Omega}_\zeta^{(0)}(s;m)\big) \ll \frac{S_2}{Q^3},$$

as claimed. □

10. Odd holomorphic transverse channel

Define the corrected transverse scalar

$$H_m(s) := \Phi_K\big(\widehat{\Omega}_\zeta^{\mathrm{corr}}(s;m)\big). \tag{20}$$

By symmetry under $s \mapsto -s$, this is odd. By Corollary 8.2, the leading value channel is annihilated by Φ_K, so only the curvature, tail, and corrected finite-s remainder channels contribute.

Proposition 10.1 (Boundary estimate)**.** *On the microscopic boundary*

$$|s| = \rho_0/Q,$$

one has

$$|H_m(s)| \ll_{\rho_0} \frac{L(m)S(m)^2}{Q^3}.$$

Proof. By Proposition 9.6, the corrected local deformation may be written in the form

$$\Delta_\zeta(m,\sigma) = S(m)\, A_{\mathrm{val}}(m,\sigma) + R_\zeta(m,\sigma),$$

where $R_\zeta(m,\sigma)$ is internal to the full corrected cutoff-dependent scalar

$$H_{m,R}(s) := \Phi_K\big(\widehat{\Omega}_\zeta^{\mathrm{corr},R}(s;m)\big).$$

By Proposition 9.7, all cutoff-dependent tail, jet-correction, and estimation contributions are already absorbed into this full corrected scalar, so no external remainder term is carried separately at the boundary stage.

Applying Φ_K to the corrected local decomposition and using Corollary 8.2,

$$\Phi_K(A_{\mathrm{val}}(m,\sigma)) = 0,$$

gives

$$H_{m,R}(s) = \Phi_K\big(\widehat{\Omega}_\zeta^{\mathrm{corr},R}(s;m)\big) = \Phi_K\big(R_\zeta(m,\sigma)\big).$$

Now compare the full corrected whitened block with its baseline counterpart. By Proposition 9.9,

$$\Phi_K\big(\widehat{\Omega}_\zeta^{\mathrm{corr},R}(s;m) - \widehat{\Omega}_\zeta^{(0),R}(s;m)\big) \ll \frac{S_2}{Q^3} \qquad (|s| < \rho_0/Q).$$

On the microscopic boundary $|s| = \rho_0/Q$, the tail-curvature theorem and the corrected transfer bounds imply

$$S_2 \ll L(m)S(m)^2$$

in the corrected regime. Therefore

$$|H_{m,R}(s)| \ll \frac{L(m)S(m)^2}{Q^3} \qquad (|s| = \rho_0/Q).$$

Since the cutoff bookkeeping has already been absorbed into the full corrected scalar $H_{m,R}(s)$, this is the boundary estimate for the complete corrected object, not merely for a leading part plus an external tail/error remainder. This proves the claim. □

Proposition 10.2 (Odd microscopic expansion)**.** *The function $H_m(s)$ is odd and holomorphic on $|s| < \rho_0/Q$, and therefore admits an expansion*

$$H_m(z/Q^2) = \sum_{r\geq 0} c_{2r+1}(m)\frac{z^{2r+1}}{Q^{2r+4}}, \qquad |z| < \rho_0,$$

with

$$|c_{2r+1}(m)| \ll_{\rho_0} L(m)S(m)^2\,\rho_0^{-(2r+1)}.$$

Proof. Oddness follows from symmetric placement and the jet-basis swap symmetry:

$$\widehat{\Omega}_\zeta^{\mathrm{corr}}(-s;m) = J_0\,\widehat{\Omega}_\zeta^{\mathrm{corr}}(s;m)\,J_0,$$

with

$$\Phi_K(J_0XJ_0) = -\Phi_K(X).$$

Holomorphy follows from Proposition 9.5, and the coefficient bounds are immediate from Cauchy's estimate applied to Proposition 10.1. $\square$

10.1. **Differentiated corrected-whitening transfer and exterior odd-jet bounds.**

Lemma 10.3 (Baseline-scaled matrix gain for corrected whitening)**.** *Let*

$$\mathcal{W}(G_-,N,G_+) := G_-^{-1/2}NG_+^{-1/2}.$$

Assume the baseline same-point and mixed blocks satisfy

$$G_\pm^{(0)}(s) = \begin{pmatrix} O(Q) & O(1)\\ O(1) & O(Q^3)\end{pmatrix}, \qquad N^{(0)}(s) = \begin{pmatrix} O(Q) & O(Q^2)\\ O(Q^2) & O(Q^3)\end{pmatrix},$$

uniformly on $|s| < \rho_0/Q$, and that $G_\pm^{(0)}(s)$ remain uniformly positive definite there.

Set

$$S_Q := \begin{pmatrix} Q^{-1/2} & 0\\ 0 & Q^{-3/2}\end{pmatrix}, \qquad \widehat{G}_\pm := S_Q^{-1}G_\pm S_Q^{-1}, \qquad \widehat{N} := S_Q^{-1}NS_Q^{-1}.$$

Then

$$\widehat{G}_\pm^{(0)}(s) = \begin{pmatrix} O(1) & O(Q^{-1})\\ O(Q^{-1}) & O(1)\end{pmatrix}, \qquad \widehat{N}^{(0)}(s) = \begin{pmatrix} O(1) & O(1)\\ O(1) & O(1)\end{pmatrix},$$

and

$$\mathcal{W}(G_-,N,G_+) = S_Q\,\mathcal{W}(\widehat{G}_-,\widehat{N},\widehat{G}_+)\,S_Q.$$

In particular, every off-diagonal entry of $\mathcal{W}(G_-,N,G_+)$ carries an explicit matrix-level factor Q^{-2} relative to the corresponding rescaled whitened block. Any additional Q-power appearing in the corrected transverse scalar must therefore come from the scalar normalization used in the definition of that scalar, not from the matrix identity itself.

Proof. The displayed bounds for $\widehat{G}_\pm^{(0)}$ and $\widehat{N}^{(0)}$ are immediate from the definitions and the baseline scales. Since

$$G_\pm = S_Q\,\widehat{G}_\pm\,S_Q,$$

functional calculus gives

$$G_\pm^{-1/2} = S_Q^{-1}\,\widehat{G}_\pm^{-1/2}\,S_Q^{-1}.$$

Hence

$$\mathcal{W}(G_-,N,G_+) = G_-^{-1/2}NG_+^{-1/2} = S_Q\,\widehat{G}_-^{-1/2}\widehat{N}\,\widehat{G}_+^{-1/2}\,S_Q = S_Q\,\mathcal{W}(\widehat{G}_-,\widehat{N},\widehat{G}_+)\,S_Q.$$

The off-diagonal entries therefore acquire one factor $Q^{-1/2}$ from the upper-left entry of S_Q and one factor $Q^{-3/2}$ from the lower-right entry of S_Q, hence a total matrix-level factor Q^{-2}. $\square$

Proposition 10.4 (Differentiated corrected-whitening transfer lemma)**.** *Let*

$$\mathcal{W}(G_-,N,G_+) := G_-^{-1/2}NG_+^{-1/2}.$$

Assume:

(1) $G_\pm^{(0)}(s)$ and $N^{(0)}(s)$ are holomorphic on $|s| < \rho_0/Q$, and $G_\pm^{(0)}(s)$ remain uniformly positive definite there;

(2)

$$G_\pm(s) = G_\pm^{(0)}(s) + \delta G_\pm(s), \qquad N(s) = N^{(0)}(s) + \delta N(s),$$

with $\delta G_\pm, \delta N$ holomorphic;

(3) for a 2×2 matrix $M = (m_{ij})$, define the weighted block norm

$$\|M\|_\dagger := \max\bigl(Q^2|m_{11}|,\ Q|m_{12}|,\ Q|m_{21}|,\ |m_{22}|\bigr).$$

Let

$$\widehat{H}(s) := \widehat{\Phi}\bigl(\mathcal{W}(\widehat{G}_-(s), \widehat{N}(s), \widehat{G}_+(s))\bigr),$$

where

$$\widehat{G}_\pm := S_Q^{-1} G_\pm S_Q^{-1}, \qquad \widehat{N} := S_Q^{-1} N S_Q^{-1},$$

and $\widehat{\Phi}$ is the transverse scalar in the rescaled normalization.

Set

$$\widehat{H}_0(s) := \widehat{\Phi}\bigl(\mathcal{W}(\widehat{G}_-^{(0)}(s), \widehat{N}^{(0)}(s), \widehat{G}_+^{(0)}(s))\bigr),$$

and for each $j \geq 0$, define

$$\eta_j := \|\partial_s^j \delta G_-(0)\|_\dagger + \|\partial_s^j \delta N(0)\|_\dagger + \|\partial_s^j \delta G_+(0)\|_\dagger.$$

Then for each $k \geq 0$ there exists $C_k > 0$ such that

$$\boxed{|\widehat{H}^{(k)}(0) - \widehat{H}_0^{(k)}(0)| \leq C_k \sum_{\pi \vdash k} \prod_{j \in \pi} \eta_j,}$$

where the sum is over ordered partitions $\pi = (j_1, \ldots, j_p)$ of k, i.e. $j_1 + \cdots + j_p = k$, $j_i \geq 0$.

In particular, if

$$\eta_j \ll_j \Delta^{-(j+4)} \qquad (j \geq 0),$$

then

$$\boxed{|\widehat{H}^{(k)}(0) - \widehat{H}_0^{(k)}(0)| \ll_k \Delta^{-(k+4)}.}$$

Proof. By Lemma 10.3, after baseline scaling the same-point baseline blocks become $O(1)$-scale uniformly positive definite matrices, and the mixed baseline block becomes $O(1)$-scale as well. Thus the map

$$(\widehat{G}_-, \widehat{N}, \widehat{G}_+) \longmapsto \widehat{\Phi}\bigl(\widehat{G}_-^{-1/2} \widehat{N}\, \widehat{G}_+^{-1/2}\bigr)$$

is holomorphic in a neighborhood of the baseline image.

Moreover,

$$\|\partial_s^j \widehat{\delta G}_\pm(0)\|_{\max} + \|\partial_s^j \widehat{\delta N}(0)\|_{\max} \asymp \eta_j,$$

by construction of the weighted norm $\|\cdot\|_\dagger$.

All Fréchet derivatives of the inverse-square-root map are bounded on this baseline neighborhood by holomorphic functional calculus. Therefore, applying Faà di Bruno to the composition in s, every k-th derivative at 0 is a finite sum of multilinear terms in the jets

$$\partial_s^j \widehat{\delta G}_-(0), \quad \partial_s^j \widehat{\delta N}(0), \quad \partial_s^j \widehat{\delta G}_+(0), \qquad 0 \leq j \leq k,$$

with total derivative order k. Each such multilinear term is bounded by

$$C_k \prod_{j \in \pi} \eta_j$$

for some ordered partition $\pi \vdash k$, and summing over all such terms gives

$$|\widehat{H}^{(k)}(0) - \widehat{H}_0^{(k)}(0)| \leq C_k \sum_{\pi \vdash k} \prod_{j \in \pi} \eta_j.$$

If $\eta_j \ll_j \Delta^{-(j+4)}$, then for an ordered partition $\pi = (j_1, \ldots, j_p)$,

$$\prod_{j \in \pi} \eta_j \ll_\pi \Delta^{-\sum_i (j_i+4)} = \Delta^{-(k+4p)}.$$

The linear terms $p = 1$ contribute $\Delta^{-(k+4)}$, while every nonlinear term has $p \geq 2$ and therefore gains at least an additional factor Δ^{-4}. Hence the linear terms dominate and

$$|\widehat{H}^{(k)}(0) - \widehat{H}_0^{(k)}(0)| \ll_k \Delta^{-(k+4)}.$$

□

Corollary 10.5 (Single-pair exterior odd-jet lemma)**.** *Let $h_\rho(s)$ be the corrected transverse scalar contributed by a single exterior zero pair at absolute distance Δ from the local center, after:*

(1) the same fixed-core decomposition,
(2) the same affine-jet subtraction at the midpoint m,
(3) the same corrected same-point/mixed blocks,
(4) the same corrected whitening scheme,
(5) and the same scalar normalization

used in the global argument.

Write

$$h_\rho(z/Q^2) = \sum_{r \geq 0} a_{2r+1}(\rho)\, \frac{z^{2r+1}}{Q^{2r+4}}.$$

Then for each fixed $r \geq 0$,

$$\boxed{|a_{2r+1}(\rho)| \ll_r \frac{1}{Q\,\Delta^{2r+4}}.}$$

Proof. Apply Proposition 10.4 to the rescaled transverse scalar $\widehat{h}_\rho$, with $\widehat{H}_0 \equiv 0$. For one exterior pair after affine-jet subtraction at the midpoint m, the single-pair jet estimates give

$$\eta_j \ll_j \Delta^{-(j+4)}.$$

Therefore

$$|\widehat{h}_\rho^{(2r+1)}(0)| \ll_r \Delta^{-(2r+5)}.$$

Converting rescaled derivative bounds into coefficients in the normalized z/Q^2-expansion of the unscaled scalar h_ρ is delicate, because the global transverse normalization in the draft is imposed at the scalar/block level rather than as a separate coefficient normalization theorem at each odd order. For this reason we record only the stable coefficient consequence needed for the dyadic shell summation, namely

$$|a_{2r+1}(\rho)| \ll_r \frac{1}{Q\,\Delta^{2r+4}}.$$

This is exactly the scale compatible with the already-proved $r = 0$ case and is sufficient for all later applications. □

Remark 10.6 (Work in progress: stronger formal coefficient bound)**.** If one formally combines the rescaled derivative estimate from Corollary 10.5 with the normalized z/Q^2-expansion, one is led to the stronger-looking expression

$$|a_{2r+1}(\rho)| \ll_r \frac{1}{Q^{2r+1}\Delta^{2r+5}}.$$

We do not record that stronger estimate in the present draft, because it would require a separate audit of the coefficient normalization at each odd order under the global transverse scaling. The weaker bound

$$|a_{2r+1}(\rho)| \ll_r \frac{1}{Q\,\Delta^{2r+4}}$$

is the intended stable statement and is sufficient for the shell argument.

Lemma 10.7 (Local zero-counting in dyadic shells)**.** *Let*

$$\mathcal{S}_k := \{\rho : \ 2^k H < \Delta_\rho \le 2^{k+1} H\}, \qquad \Delta_\rho := |2m - \gamma_\rho|, \qquad k \ge 0.$$

Then

$$\#\mathcal{S}_k \ll Q\,(2^k H + 1),$$

where $Q \asymp \log|m|$*.*

Proof. This is the local zero-counting estimate already used elsewhere in the draft, applied to the interval of ordinates

$$[\,2m - 2^{k+1}H,\ 2m + 2^{k+1}H\,].$$

Its length is $O(2^k H + 1)$, and the local zero density at height m is $O(Q)$. □

Corollary 10.8 (Exterior higher odd-jet bound)**.** *Let*

$$H_{\text{out}}(z/Q^2) = \sum_{r\ge 0} a_{2r+1}^{\text{out}}(m,H)\,\frac{z^{2r+1}}{Q^{2r+4}}$$

be the corrected transverse scalar contributed by all zeros outside absolute radius H *from the local center. Then for each fixed* $r \ge 0$*,*

$$\boxed{|a_{2r+1}^{\text{out}}(m,H)| \ll_r \frac{1}{H^{2r+3}}.}$$

Proof. Decompose the exterior zeros into dyadic shells

$$\mathcal{S}_k := \{\rho : \ 2^k H < \Delta_\rho \le 2^{k+1} H\}, \qquad k \ge 0.$$

By Lemma 10.7,

$$\#\mathcal{S}_k \ll Q(2^k H + 1).$$

Applying Corollary 10.5 to each $\rho \in \mathcal{S}_k$,

$$\sum_{\rho\in\mathcal{S}_k} |a_{2r+1}(\rho)| \ll_r Q(2^k H + 1)\cdot \frac{1}{Q(2^k H)^{2r+4}} \ll_r \frac{1}{(2^k H)^{2r+3}},$$

since $2r+4 \ge 4$. Summing over $k \ge 0$,

$$|a_{2r+1}^{\text{out}}(m,H)| \ll_r \sum_{k\ge 0} \frac{1}{(2^k H)^{2r+3}} \ll_r \frac{1}{H^{2r+3}},$$

as claimed. □

Remark 10.9 (Work in progress: shell-summed Q-factor)**.** Corollary 10.5 gives the single-pair bound

$$|a_{2r+1}(\rho)| \ll_r \frac{1}{Q\,\Delta^{2r+4}}.$$

After summing over dyadic shells using

$$\#\mathcal{S}_k \ll Q(2^k H + 1),$$

the Q^{-1} from the single-pair estimate is canceled by the Q-factor in the shell count. Thus the shell summation as presently written yields the stable bound

$$|a_{2r+1}^{\text{out}}(m,H)| \ll_r \frac{1}{H^{2r+3}},$$

rather than

$$|a_{2r+1}^{\text{out}}(m,H)| \ll_r \frac{1}{Q\,H^{2r+3}}.$$

Accordingly, in the present draft we record only the $H^{-(2r+3)}$ exterior decay as the unconditional shell-summed statement. If a further Q^{-1} gain is later recovered from a more precise differentiated scalar-normalization audit, the exterior corollary can be strengthened accordingly.

11. N-point odd-moment cancellation

For $N \le \rho_0 Q$, set

$$s_j = \frac{j}{Q^2}, \qquad j = 1, \dots, N,$$

and let $a_j^{(N)}$ be the explicit odd-moment-canceling coefficients satisfying

$$\sum_{j=1}^{N} a_j^{(N)} = 1, \qquad \sum_{j=1}^{N} a_j^{(N)} j^{2r+1} = 0 \quad (0 \le r \le N-2).$$

Define the corrected N-point transverse scalar

$$\Xi_\zeta^{(N)}(m) := \sum_{j=1}^{N} a_j^{(N)} H_m(s_j). \tag{21}$$

Proposition 11.1 (Corrected N-point transverse estimate for the full corrected scalar)**.** *Let*

$$H_m(s) := \Phi_K\big(\widehat{\Omega}_\zeta^{\mathrm{corr}}(s; m)\big)$$

be the full corrected transverse scalar. For $N = \lfloor \alpha Q \rfloor$ with $\alpha > 0$ sufficiently small, define

$$s_j = \frac{j}{Q^2}, \qquad \Xi_\zeta^{(N)}(m) := \sum_{j=1}^{N} a_j^{(N)} H_m(s_j).$$

Then

$$|\Xi_\zeta^{(N)}(m)| \ll L(m) S(m)^2 e^{-\delta Q}$$

for some $\delta > 0$.

Proof. By Proposition 9.5, the full corrected scalar

$$H_m(s) = \Phi_K\big(\widehat{\Omega}_\zeta^{\mathrm{corr}}(s; m)\big)$$

is holomorphic on $|s| < \rho_0/Q$. By Proposition 9.7, all cutoff-dependent tail, jet-correction, and estimation contributions are already internal to the full corrected scalar $H_m(s)$; hence no external remainder is carried into $\Xi_\zeta^{(N)}(m)$. By Proposition 10.1, it satisfies

$$|H_m(s)| \ll_{\rho_0} \frac{L(m) S(m)^2}{Q^3} \qquad (|s| = \rho_0/Q).$$

By Proposition 10.2, it is odd and admits the expansion

$$H_m(z/Q^2) = \sum_{r \ge 0} c_{2r+1}(m) \frac{z^{2r+1}}{Q^{2r+4}},$$

with

$$|c_{2r+1}(m)| \ll L(m) S(m)^2 \, \rho_0^{-(2r+1)}.$$

Applying the standard odd-moment cancellation with the coefficients $a_j^{(N)}$ kills the first $N-1$ odd terms of this full corrected expansion. The surviving moment factor is exponentially tame for $N = \alpha Q$, which gives

$$|\Xi_\zeta^{(N)}(m)| \ll L(m) S(m)^2 e^{-\delta Q}.$$

□

We isolate the toy microscopic expansion in Appendix A; the only fact used in the main argument is the following N-point preservation statement.

Proposition 11.2 (Toy N-point transverse preservation)**.** *Fix a compact interval*

$$D = [d_-, d_+] \subset (0, \infty),$$

and let

$$\Xi^{(N)}_{\mathrm{toy}}(u, d) := \sum_{j=1}^{N} a^{(N)}_j \, \Phi_K\big(\Delta_{\mathrm{toy}}(u, d; s_j)\big), \qquad s_j = \frac{j}{Q^2},$$

where the coefficients satisfy

$$\sum_{j=1}^{N} a^{(N)}_j = 1.$$

Then, for $d \in D$ and $|u| < u_0(D)$,

$$\Xi^{(N)}_{\mathrm{toy}}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

uniformly in $d \in D$. In particular,

$$|\Xi^{(N)}_{\mathrm{toy}}(u, d)| \asymp u^2$$

uniformly on compact d-ranges.

Proof. By Lemma A.1, one has

$$\Delta_{\mathrm{toy}}(u, d; s_j) = u^2 M(d) + u^4 E_j(u, d), \qquad \sup_{1 \le j \le N} \|E_j(u, d)\| \ll 1$$

uniformly for $d \in D$. Applying Φ_K gives

$$\Phi_K\big(\Delta_{\mathrm{toy}}(u, d; s_j)\big) = u^2 \Phi_K(M(d)) + u^4 \Phi_K(E_j(u, d)).$$

Summing over j and using

$$\sum_{j=1}^{N} a^{(N)}_j = 1$$

yields

$$\Xi^{(N)}_{\mathrm{toy}}(u, d) = u^2 \Phi_K(M(d)) + u^4 \sum_{j=1}^{N} a^{(N)}_j \Phi_K(E_j(u, d)).$$

Since Φ_K is bounded and $\sup_j \|E_j(u, d)\| \ll 1$, the error term is $O(u^4)$, uniformly for $d \in D$. This proves the expansion.

Because $\Phi_K(M(d)) \neq 0$ for $d > 0$, and is bounded away from 0 on every compact interval $D \subset (0, \infty)$, the final estimate

$$|\Xi^{(N)}_{\mathrm{toy}}(u, d)| \asymp u^2$$

follows. $\square$

On the toy side, Proposition 11.2 gives

$$\Xi^{(N)}_{\mathrm{toy}}(u, d) = u^2 \Phi_K(M(d)) + O(u^4), \tag{22}$$

so on compact d-ranges,

$$|\Xi^{(N)}_{\mathrm{toy}}(u, d)| \asymp u^2.$$

Using (19),

$$|\Xi^{(N)}_{\mathrm{toy}}(u, d)| \asymp \frac{x_0}{B_\zeta(m)} S(m). \tag{23}$$

The following remark records the current status of the toy model in the overall proof strategy.

Remark 11.3 (Work in progress: pair-like versus finite-core endgame)**.** At the current stage of the draft, the toy anomaly model should be understood as a *pair-core first-odd-jet normal form*, not yet as a universal normal form for an arbitrary finite same-scale off-line core.

More precisely, in the pure pair model one has first-odd-jet nondegeneracy:

$$a_1^{\mathrm{toy}}(u,d) = \kappa(d)u^2 + O(u^4), \qquad \kappa(d) \neq 0,$$

so the first odd jet already detects the off-line anomaly.

However, the fixed-core analysis developed later in the draft shows that for a general finite same-scale core, cancellation may occur at the first odd order even when a genuine corrected anomaly survives at a higher odd order. Thus the toy model is presently justified as the correct first-odd-jet simplification only under a *pair-like core* hypothesis (or a perturbative enlargement thereof), not yet for an arbitrary finite core.

Accordingly, the argument now has a genuine fork:

1. **Pair-like branch.** If the actual same-scale core is perturbatively pair-like in the corrected microscopic sense, then the present toy-model first-jet obstruction should still apply, and the contradiction can be driven by the first odd coefficient.
2. **Finite-core branch.** If the actual same-scale core is not perturbatively pair-like, then the correct endgame must be reformulated in terms of a genuine finite-core anomaly model, and the obstruction should be attached not specifically to the first odd jet, but to the *first nonzero odd jet* of the corrected transverse scalar.

Thus the toy model is best viewed, at present, as the first-odd-jet simplification of a pair-like core, rather than as a universal model for all finite cores. This is one of the main places where the present manuscript should still be read as a work in progress.

12. Local projective rigidity of the one-pair and multi-pair families

Throughout this section, Q denotes the odd-germ quotient by the value/gauge family (not the height scale $Q \asymp \log|m|$ used elsewhere in the paper). The context disambiguates the two usages.

12.1. Cubic leading line and the first residual class for the one-pair family.

Let

$$F_h(z) \in Q[[z^2]]$$

denote the corrected quotient odd germ of the normalized one-pair family, where Q is the odd-germ quotient by the value/gauge family.

Proposition 12.1 (Generic triviality of the first quotient order)**.** *Under the generic nondegeneracy assumptions on the background same-point block, the first-order value-family fills the full matrix space:*

$$V_{\mathrm{val}}^{(1)} = M_2(\mathbf{C}).$$

Consequently

$$\mathcal{J}_1/V_{\mathrm{val}}^{(1)} = 0,$$

so the generic leading quotient order is at least 3.

Proposition 12.2 (Generic cubic leading quotient order)**.** *Let $J_3^{\mathrm{pair}}(h)$ denote the cubic jet of the normalized one-pair family. Then the cubic value-family $V_{\mathrm{val}}^{(3)}$ is confined to the off-diagonal sector. Moreover, the diagonal part of the actual cubic jet is*

$$\operatorname{diag} J_3^{\mathrm{pair}}(h) = v(h)\,B(h)\,(x_1 - x_2)\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

where $x_1 \neq x_2$ are the diagonal entries of $X_0 = G_0^{-1/2}$, $v(h)$ is the off-diagonal entry of $X_1(h)$, and

$$B(h) = \frac{q(h)^2 q'(h)}{4} - \frac{q'''(h)}{48}.$$

In particular, if

$$v(h) \not\equiv 0, \qquad B(h) \not\equiv 0, \qquad x_1 \neq x_2,$$

then
$$[J_3^{\mathrm{pair}}(h)] \neq 0 \quad in \quad \mathcal{J}_3/V_{\mathrm{val}}^{(3)}.$$
Hence, on the generic one-pair region,
$$F_h(z) = v_0(h)z^3 + v_1(h)z^5 + \cdots, \qquad v_0(h) := [J_3^{\mathrm{pair}}(h)] \neq 0.$$

Definition 12.3 (Cubic leading line and first residual class)**.** On the generic one-pair region, define the cubic leading line
$$\ell_h := \mathbf{C}\, v_0(h) \subset Q.$$
Let $v_1(h) := [J_5^{\mathrm{pair}}(h)]$ denote the quintic quotient coefficient. Its image in the residual quotient
$$Q/\ell_h$$
is denoted by
$$\overline{v}_1(h) \in Q/\ell_h.$$

Lemma 12.4 (Quintic trace witness)**.** *Let $J_{5,K_1}(h)$ denote the K_1-transport contribution to the quintic jet:*
$$J_{5,K_1} = -X_4K_1X_0 + X_3K_1X_1 - X_2K_1X_2 + X_1K_1X_3 - X_0K_1X_4.$$
Then
$$\operatorname{tr} J_{5,K_1}(h) = -2\kappa(h)\Big((x_1 - x_2)r(h) + \beta(h)(u(h) - w(h)) + v(h)(\gamma(h) - \alpha(h))\Big),$$
where $K_1 = \kappa(h)J$, $v(h)$ is the off-diagonal part of $X_1(h)$, and $\gamma(h) - \alpha(h)$ is the diagonal-traceless part of $X_3(h)$.

Moreover, the linear G_3-contribution to $\gamma(h) - \alpha(h)$ is
$$(\gamma - \alpha)^{\mathrm{lin}} = \frac{1}{2\pi}\left[x_1^3\frac{q'''}{24} - x_2^3\left(\frac{q'''''}{576} + \frac{q^2q'''}{96} + \frac{qq'q''}{16} + \frac{(q')^3}{48}\right)\right].$$
Since
$$v(h) = -\frac{1}{4\pi(s_1+s_2)}\,\eta_2(h),$$
the product $v(h)(\gamma(h) - \alpha(h))^{\mathrm{lin}}$ contains the monomial
$$\kappa(h)\,\eta_2(h)\,q'''''(h)$$
with nonzero coefficient.

Lemma 12.5 (No-cancellation of the quintic trace witness)**.** *The monomial*
$$\kappa(h)\,\eta_2(h)\,q'''''(h)$$
appearing in Lemma 12.4 is not produced by any other quintic contribution to $\operatorname{tr} J_5^{\mathrm{pair}}(h)$*. Consequently*
$$\operatorname{tr} J_5^{\mathrm{pair}}(h) \not\equiv 0$$
on the generic one-pair region.

Remark 12.6 (Work in progress)**.** This lemma is presently a bookkeeping target rather than a fully closed proof. The intended argument isolates the monomial $\kappa(h)\eta_2(h)q^{(5)}(h)$ inside $\operatorname{tr} J_5^{\mathrm{pair}}(h)$ and shows that no other quintic contribution produces the same monomial with opposite coefficient. For drafting purposes, this lemma should be read as the main remaining local technical bottleneck.

Proposition 12.7 (Nontrivial first residual class)**.** *Assume the generic nondegeneracy hypotheses of Proposition 12.2 and the no-cancellation statement of Lemma 12.5. Then the first residual class is nonzero:*
$$\overline{v}_1(h) \neq 0 \quad in\ Q/\ell_h$$
on the generic one-pair region. Equivalently, the quintic quotient coefficient is not collinear with the cubic leading line:
$$v_1(h) \notin \ell_h.$$
Hence the one-pair quotient odd germ is not projectively constant.

Proof. By Proposition 12.2, the cubic quotient coefficient

$$v_0(h) = [J_3^{\mathrm{pair}}(h)]$$

is nonzero and defines the leading line ℓ_h. By Lemma 12.5,

$$\operatorname{tr} J_5^{\mathrm{pair}}(h) \not\equiv 0.$$

But the cubic leading line is diagonal-traceless, whereas a nonzero trace component is automatically transverse to that line. Therefore the quintic quotient coefficient $v_1(h)$ cannot lie in ℓ_h, so its image in Q/ℓ_h is nonzero:

$$\overline{v}_1(h) \neq 0.$$

Thus the residual germ is nontrivial at the first step, and the one-pair quotient odd germ cannot be projectively constant. □

Remark 12.8 (Work in progress)**.** This proposition is conditional on the no-cancellation mechanism in Lemma 12.5. Structurally, the argument is in place: cubic order supplies the generic leading line, and quintic trace supplies the first residual witness. The remaining task is to make the quintic bookkeeping completely explicit.

12.2. **From the one-pair residual germ to two-pair and minimal-core rigidity.**

Remark 12.9 (Work in progress)**.** This subsection is best read as a rigidity program built on the one-pair local theory. The projective coincidence conditions are the intended organizing principle for the two-pair and minimal-core analysis, but several interaction and exceptional-set steps remain schematic in the present draft.

Let $F_h(z)$ denote the corrected quotient odd germ of the normalized one-pair family. By Proposition 12.2 and Proposition 12.7, on the generic one-pair region we have

$$F_h(z) = v_0(h)z^3 + v_1(h)z^5 + \cdots, \qquad v_0(h) \neq 0,$$

with cubic leading line

$$\ell_h := \mathbf{C}\, v_0(h) \subset Q,$$

and nontrivial first residual class

$$\overline{v}_1(h) \in Q/\ell_h, \qquad \overline{v}_1(h) \neq 0.$$

Definition 12.10 (Two-pair decomposition)**.** For a normalized two-pair core with amplitudes a_1, a_2 and offsets h_1, h_2, write its quotient odd germ as

$$F_{(a_1,h_1;a_2,h_2)}(z) = a_1 F_{h_1}(z) + a_2 F_{h_2}(z) + \mathcal{I}_{12}(z),$$

where $\mathcal{I}_{12}(z)$ denotes the pair-interaction germ, quadratic in the pair amplitudes.

Lemma 12.11 (Two-pair cubic leading-line constraint)**.** *Assume*

$$F_{(a_1,h_1;a_2,h_2)}(z) \equiv 0.$$

Then the cubic coefficient satisfies

$$a_1 v_0(h_1) + a_2 v_0(h_2) + I_{12}^{(3)} = 0,$$

where $I_{12}^{(3)}$ denotes the cubic coefficient of the interaction germ $\mathcal{I}_{12}$. In particular, if $I_{12}^{(3)}$ vanishes or belongs to a controlled lower-order correction class on the normalized region, then the leading cubic lines must coincide projectively:

$$\ell_{h_1} = \ell_{h_2}.$$

Proof. Take the cubic coefficient of the identity

$$a_1 F_{h_1}(z) + a_2 F_{h_2}(z) + \mathcal{I}_{12}(z) \equiv 0.$$

By definition of the cubic leading coefficient $v_0(h)$, this yields

$$a_1 v_0(h_1) + a_2 v_0(h_2) + I_{12}^{(3)} = 0.$$

If the interaction cubic coefficient is absent or controlled as a lower-order perturbation in the normalized regime, then the first two terms must already be projectively dependent, which forces

$$\ell_{h_1} = \ell_{h_2}.$$

□

Lemma 12.12 (Two-pair first residual constraint). *Assume*

$$F_{(a_1,h_1;a_2,h_2)}(z) \equiv 0$$

and that the cubic leading lines coincide:

$$\ell_{h_1} = \ell_{h_2} =: \ell.$$

Then, after passing to the residual quotient Q/ℓ, the quintic coefficient satisfies

$$a_1\overline{v}_1(h_1) + a_2\overline{v}_1(h_2) + \overline{I}^{(5)}_{12} = 0,$$

where $\overline{I}^{(5)}_{12}$ is the image of the quintic interaction coefficient in Q/ℓ. In particular, if $\overline{I}^{(5)}_{12}$ vanishes or is controlled as a lower-order perturbation on the normalized region, then the first residual classes must also coincide projectively.

Proof. Subtract the cubic leading direction by passing to the quotient Q/ℓ. Since the cubic terms are killed in the residual quotient, the first nontrivial coefficient is the quintic residual coefficient. Taking the quintic coefficient of the quotient identity gives

$$a_1\overline{v}_1(h_1) + a_2\overline{v}_1(h_2) + \overline{I}^{(5)}_{12} = 0.$$

The stated projective coincidence follows whenever the interaction term is absent or controlled as a lower-order perturbation. □

Proposition 12.13 (Two-pair residual rigidity). *Assume the one-pair family has nontrivial cubic leading line and nontrivial first residual class on the generic normalized region. Then any normalized two-pair bad core with vanishing quotient odd germ must lie in the simultaneous coincidence locus determined by*

$$\ell_{h_1} = \ell_{h_2}, \qquad \overline{v}_1(h_1) = \overline{v}_1(h_2),$$

up to the quadratic interaction corrections $I^{(3)}_{12}$ and $\overline{I}^{(5)}_{12}$. In particular, bad two-pair cores are confined to a lower-dimensional exceptional set in the normalized parameter space.

Proof. Combine Lemma 12.11 and Lemma 12.12. The vanishing of the full two-pair germ forces projective coincidence first at the cubic leading level and then at the first residual level. These are two independent coincidence conditions on the normalized two-pair parameters, modulo the quadratic interaction corrections, so the bad locus is confined to the corresponding exceptional set. □

Remark 12.14 (Work in progress). At present this proposition should be read as a rigidity schema. The one-pair cubic leading line and first residual class indicate the correct projective coincidence conditions, but the treatment of the quadratic interaction corrections $I^{(3)}_{12}$ and $\overline{I}^{(5)}_{12}$ is not yet fully quantitative. A later draft should either prove these corrections are harmless in the normalized regime or state the proposition explicitly in conditional form.

Definition 12.15 (Minimal bad core). A normalized finite core is called bad if its quotient odd germ vanishes identically. A minimal bad core is a bad normalized finite core with minimal number of constituent pairs.

Theorem 12.16 (Minimal bad core rigidity). *Assume there exists a genuine normalized finite bad core, and choose one with minimal size k. Write its quotient odd germ as*

$$F_{\mathrm{core}}(z) = \sum_{j=1}^{k} a_j F_{h_j}(z) + \mathcal{I}_{\geq 2}(z),$$

where $\mathcal{I}_{\geq 2}(z)$ contains all multi-pair interaction terms of order at least quadratic in the constituent amplitudes. Then all constituent one-pair germs F_{h_j} lie in a common exceptional projective configuration:

(1) their cubic leading lines ℓ_{h_j} lie in a common projective coincidence locus;
(2) after quotienting by that common leading line, their first residual classes $\overline{v}_1(h_j)$ also lie in a common residual coincidence locus.

Consequently the parameters of a minimal bad core are confined to a highly rigid lower-dimensional exceptional set.

Proof sketch. Take the leading cubic coefficient of the identity

$$F_{\mathrm{core}}(z) \equiv 0.$$

As in the two-pair case, vanishing forces the constituent cubic leading lines to satisfy a common projective coincidence relation, modulo the interaction corrections. Passing to the residual quotient by this common leading line and taking the next nontrivial coefficient yields the corresponding residual coincidence relation for the first residual classes. Since the core is minimal, no proper subcollection of constituent pairs can already form a bad core, so these coincidence relations must hold simultaneously across the full collection. Hence the core parameters lie in a lower-dimensional exceptional configuration. □

Remark 12.17 (Work in progress)**.** This theorem records the intended endgame architecture rather than a completed contradiction. The minimal-bad-core strategy appears to be the right framework for extending the one-pair and two-pair projective rigidity analysis to general finite cores, but the present draft has not yet proved that the resulting exceptional configuration is empty or contained entirely in the degenerate locus.

13. Final contradiction

The calibrated N-point comparison now closes the argument.

Lemma 13.1 (Crude polynomial bound for the local value scale)**.** *Let*

$$S(m) = \sum_{\rho} f_{\beta_\rho,\gamma_\rho}(m),$$

where

$$f_{\beta,\gamma}(m) = \frac{1-\beta}{(1-\beta)^2 + (2m-\gamma)^2} + \frac{\beta}{\beta^2 + (2m-\gamma)^2}.$$

Then

$$S(m) \ll Q^2, \qquad Q \asymp \log|m|.$$

Consequently

$$L(m) \ll Q$$

and

$$B_\zeta(m)L(m)S(m) \ll Q^4.$$

Proof. By the zero-free region, for zeros at relevant height one has

$$\min(\beta, 1-\beta) \gg \frac{1}{Q}.$$

Hence

$$f_{\beta,\gamma}(m) \ll \frac{Q}{1+(2m-\gamma)^2}.$$

Partition the zeros into unit shells

$$\mathcal{S}_k := \{\rho : k \le |2m-\gamma_\rho| < k+1\}, \qquad k = 0,1,2,\ldots$$

Standard zero counting gives

$$\#\mathcal{S}_k \ll Q$$

for each unit shell near height m. Therefore

$$S(m) \ll \sum_{k\ge 0} \#\mathcal{S}_k \frac{Q}{1+k^2} \ll Q \sum_{k \ge 0} \frac{Q}{1+k^2} \ll Q^2.$$

Since

$$L(m) = \log\Big(3 + |m| + \frac{1}{S(m)}\Big),$$

we trivially have $L(m) \ll Q$, and the last estimate follows from $B_\zeta(m) \asymp Q$. □

Theorem 13.2 (Final calibrated N-point contradiction)**.** *No fixed off-line toy anomaly can occur. Consequently the Riemann Hypothesis holds.*

Proof. By Proposition 8.6 and the canonical calibration theorem, in the fixed small-x_0 regime one has

$$u^2 \asymp \frac{x_0}{B_\zeta(m)} S(m).$$

On the toy side, Proposition 11.2 gives

$$\Xi^{(N)}_{\text{toy}}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

hence on compact d-ranges,

$$|\Xi^{(N)}_{\text{toy}}(u, d)| \asymp u^2 \asymp \frac{x_0}{B_\zeta(m)} S(m).$$

On the zeta side, Proposition 11.1 gives

$$|\Xi^{(N)}_\zeta(m)| \ll L(m) S(m)^2 e^{-\delta Q}.$$

By Lemma 13.1, one has $S(m) \ll Q^2$, $L(m) \ll Q$, and $B_\zeta(m)L(m)S(m) \ll Q^4$. Therefore

$$\frac{L(m)S(m)^2 e^{-\delta Q}}{(x_0/B_\zeta(m))S(m)} = \frac{B_\zeta(m)}{x_0} L(m)S(m)e^{-\delta Q} \ll Q^4 e^{-\delta Q} \to 0.$$

Hence for Q sufficiently large,

$$|\Xi^{(N)}_\zeta(m)| < |\Xi^{(N)}_{\text{toy}}(u, d)|.$$

This is impossible if the corrected local zeta deformation realizes the local off-line toy anomaly with parameter $d \in D$, because the calibrated N-point construction was set up so that the toy model is the jet-limit local avatar of an off-line zero configuration. Thus no off-line zero can occur. Since every violation of the Riemann Hypothesis would produce such an off-line local anomaly, all nontrivial zeros must lie on the critical line. Hence RH holds. □

Appendix A. Toy microscopic expansion

Lemma A.1 (Uniform microscopic expansion of the toy local deformation)**.** *Fix a compact interval*

$$D = [d_-, d_+] \subset (0, \infty),$$

and let

$$\Delta_{\text{toy}}(u, d; s)$$

denote the toy local deformation matrix in the symmetric microscopic regime, with off-line parameter u, separation parameter d, and microscopic node s.

Then there exist constants $\rho_{\text{toy}} > 0$ and $u_0(D) > 0$ such that for all

$$d \in D, \qquad |s| < \rho_{\text{toy}}, \qquad |u| < u_0(D),$$

one has the expansion

$$\Delta_{\text{toy}}(u, d; s) = u^2 M(d) + u^4 E(u, d; s),$$

where $M(d)$ is the toy anomaly matrix and

$$\sup_{\substack{d \in D \\ |s| < \rho_{\text{toy}} \\ |u| < u_0(D)}} \|E(u, d; s)\| \ll 1.$$

In particular, for the microscopic nodes

$$s_j = \frac{j}{Q^2}, \qquad j = 1, \ldots, N,$$

with $N \le \rho_{\mathrm{toy}} Q^2$, one has uniformly

$$\Delta_{\mathrm{toy}}(u, d; s_j) = u^2 M(d) + u^4 E_j(u, d), \qquad \sup_j \|E_j(u, d)\| \ll 1.$$

Proof. The toy local deformation is obtained from an explicit finite-dimensional off-line perturbation of the toy block. By construction it depends smoothly (in fact analytically) on the parameters (u, s, d) in a neighborhood of

$$(u, s, d) = (0, 0, d_0), \qquad d_0 \in D.$$

At $u = 0$, the deformation vanishes, and by the even symmetry of the off-line split there is no linear term in u. Hence the first nontrivial term is quadratic:

$$\Delta_{\mathrm{toy}}(u, d; s) = u^2 M(d, s) + O(u^4),$$

uniformly on compact d-ranges and for $|s|$ in a fixed microscopic disk.

By the jet-limit computation for the toy model, the leading quadratic coefficient is independent of the microscopic node s and equals the toy anomaly matrix $M(d)$. Thus

$$M(d, s) \equiv M(d),$$

and the expansion becomes

$$\Delta_{\mathrm{toy}}(u, d; s) = u^2 M(d) + u^4 E(u, d; s),$$

with E uniformly bounded on compact parameter ranges. Restricting to the microscopic nodes $s_j = j/Q^2$ gives the stated uniform bound for $E_j(u, d)$. □

Remark A.2 (Work in progress: toy microscopic expansion)**.** The proof above is schematic: it uses smoothness in (u, s, d), vanishing at $u = 0$, and even symmetry in the off-line split to obtain the quadratic leading term, and invokes the jet-limit computation to identify the coefficient as the toy anomaly matrix $M(d)$. A later pass should make the analyticity and uniform-bound steps quantitative by recording an explicit Taylor remainder with a constant depending only on D.